Introduction to Online Learning

Sequential decisions with convex losses

A self-contained introduction built around one running example:
forecasting the daily demand of one product as data arrive.

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1 Why online learning? A sequential forecasting story

1.1 Running example: tomorrow's demand must be predicted today

Online learning aims to predict the next value y_t as accurately as possible using only the observations $(y_1, \dots, y_{t-1})$ available so far, without requiring a probabilistic model for the sequence.

Consider one product sold by one supermarket. Each morning, before the day's demand is known, a forecaster outputs a prediction $\widehat{y}_t$. At the end of the day, demand y_t is observed and the prediction can be evaluated. The next prediction may then use all observations available so far.

This chronology is the defining feature of online learning. It is not enough to fit a model to a completed dataset and then inspect its in-sample error. The prediction at time t must be a function only of information available before y_t is revealed.

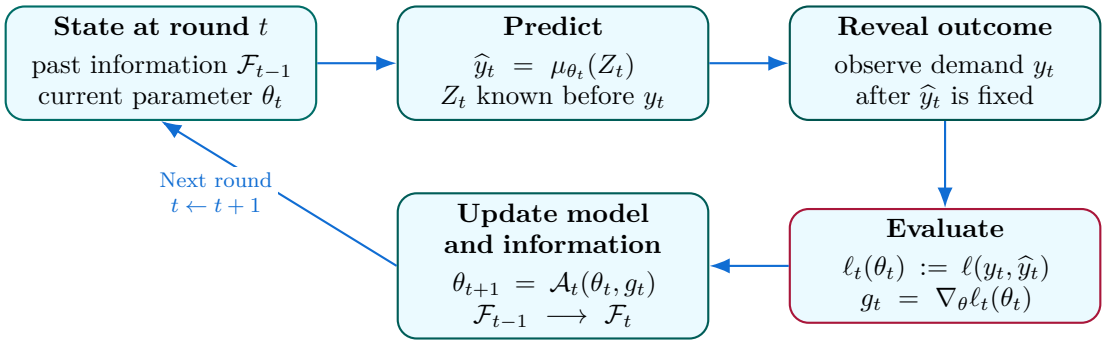


Figure 1: Chronology of one online-learning round. The input Z_t is available before demand and is $\mathcal{F}_{t-1}$-measurable. After observing y_t, the learner evaluates the loss and updates both its parameter and its information. On the next round, the updated pair $(\mathcal{F}_t, \theta_{t+1})$ becomes the new current state. The notation $\mathcal{A}_t$ denotes a generic first-order update; projected OGD later takes $\mathcal{A}_t(\theta_t, g_t) = \Pi_{\mathcal{K}}(\theta_t - \eta_t g_t)$.

The loss converts a prediction error into a numerical cost. For example, $\ell_t(\widehat{y}) = \frac{1}{2}(\widehat{y} - y_t)^2$ is the squared loss. The factor $1/2$ has no statistical meaning; it simply removes a factor 2 from the derivative.

1.2 Why not retrain a batch model every day?

A batch procedure receives a fixed dataset $(x_1, y_1), \dots, (x_n, y_n)$ and approximately solves one optimization problem. An online procedure instead interleaves prediction, observation, and updating. This distinction matters when the data stream is large, storage is limited, the environment changes, or decisions must be made immediately. Online learning also provides guarantees for arbitrary deterministic sequences; independence, stationarity, and even the existence of a data-generating distribution are not required [5, 7].

This does *not* mean that stochastic assumptions are useless. They can yield sharper statistical conclusions. It means that the basic online guarantee remains meaningful even if the sequence drifts or reacts to the learner.

The central mathematical difficulty

At round t, the learner must choose θ_t before seeing ℓ_t. If the learner knew all losses $\ell_1, \dots, \ell_T$ in advance, it could directly choose the best parameter for their sum. Online learning asks how close one can come to that hindsight benchmark while respecting the chronology.

1.3 A first update rule: Online Gradient Descent

Before introducing performance criteria or comparing several algorithms, it is useful to understand the basic mechanism by which an online learner can improve its decisions. Online Gradient Descent updates the current parameter using the local slope of the loss that has just been observed.

Definition 1.1 (Learning rate and Online Gradient Descent update)**.** After playing θ_t and observing the loss ℓ_t, the learner selects a gradient or subgradient $g_t \in \partial \ell_t(\theta_t)$. The *learning rate* $\eta_t > 0$ determines how strongly the learner reacts to this new information. The unconstrained Online Gradient Descent update is

$$\theta_{t+1} = \theta_t - \eta_t g_t. \tag{1}$$

When ℓ_t is differentiable at θ_t, this becomes

$$\theta_{t+1} = \theta_t - \eta_t \nabla \ell_t(\theta_t).$$

The negative gradient points in a direction of local decrease of the current loss (see Figure 2 for illustration). The learning rate controls the distance travelled in that direction. A very small learning rate produces slow adaptation, whereas a very large one may cause unstable updates. If decisions are constrained, the tentative update must additionally be projected onto the feasible decision set; this version will be developed in Subsection 4.3.

Before considering genuinely changing online losses, the following stationary special case explains why the update can approach an optimum.

Theorem 1.1 (Convergence in the stationary strongly convex case)**.** *Suppose that the same differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is observed at every round, so that $\ell_t = f$. Assume that f is μ-strongly convex and L-smooth* (for both notions, see Definition 2.4)*, with $0 < \mu \leq L$. Then f has a* unique minimizer $\theta^\star \in \mathbb{R}^d$*.*
For a constant learning rate satisfying $0 < \eta \leq 1/L$, the gradient update

$$\theta_{t+1} = \theta_t - \eta \nabla f(\theta_t)$$

satisfies

$$f(\theta_t) - f(\theta^\star) \leq (1 - \mu\eta)^{t-1}\big(f(\theta_1) - f(\theta^\star)\big). \tag{2}$$

Consequently, $\theta_t \xrightarrow[t\to+\infty]{} \theta^\star$ at a geometric rate.

Proof. By L-smoothness and the definition of the update *(see proof of Proposition 10.2)*,

$$f(\theta_{t+1}) \leq f(\theta_t) - \eta\left(1 - \frac{L\eta}{2}\right)\|\nabla f(\theta_t)\|_2^2.$$

Since $\eta \leq 1/L$, it follows that

$$f(\theta_{t+1}) \leq f(\theta_t) - \frac{\eta}{2}\|\nabla f(\theta_t)\|_2^2.$$

Strong convexity implies *(see proof of Proposition 10.3)*

$$\|\nabla f(\theta_t)\|_2^2 \geq 2\mu\big(f(\theta_t) - f(\theta^\star)\big).$$

Combining the two previous inequalities gives

$$f(\theta_{t+1}) - f(\theta^\star) \leq (1 - \mu\eta)\big(f(\theta_t) - f(\theta^\star)\big).$$

Iterating this inequality proves the announced bound, as written in Eq. (2). Strong convexity also gives *(see proof of Proposition 10.3)*

$$f(\theta_t) - f(\theta^\star) \geq \frac{\mu}{2}\,\|\theta_t - \theta^\star\|_2^2\,,$$

so convergence of the function values implies $\theta_t \xrightarrow[t\to+\infty]{} \theta^\star$. □

Example 1.1 (Learning a fixed demand level)**.** Suppose that the observed demand is always $y^{\text{true}} = 14$ and consider $f : \theta \mapsto \frac{1}{2}(\theta - 14)^2$. Its gradient is $f'(\theta) = \theta - 14$. Starting from $\theta_1 = 10$ with learning rate $\eta = 0.25$, gradient descent gives

$$\theta_{t+1} = \theta_t - 0.25(\theta_t - 14) = 0.75\theta_t + 0.25 \times 14.$$

By induction,

$$\theta_t - 14 = 0.75^{\,t-1}(\theta_1 - 14),$$

and therefore $\theta_t \xrightarrow[t\to+\infty]{} 14$.

At each round, the learner removes one quarter of its remaining error.

In a genuine online problem, the losses ℓ_t and observations y_t may change from one round to the next. There may then be no single optimum toward which the iterates converge. For squared loss, the update becomes

$$\theta_{t+1} = (1 - \eta_t)\theta_t + \eta_t y_t,$$

so the learner continuously tracks newly observed demand. Thus, online learning will later be evaluated through cumulative performance and regret rather than through convergence to a fixed minimizer alone.

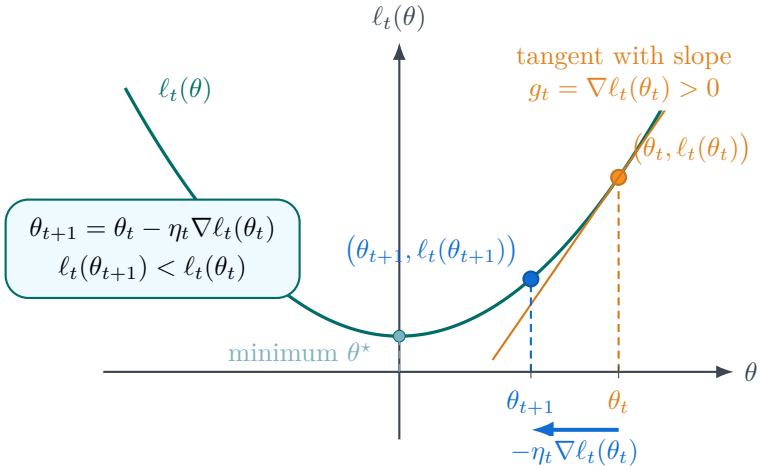


Figure 2: Illustration of one Online Gradient Descent step for a differentiable convex loss. At the current iterate θ_t, the gradient $\nabla\ell_t(\theta_t)$ is positive. The update therefore moves in the opposite direction by an amount $\eta_t\nabla\ell_t(\theta_t)$. For a suitable learning rate, the new iterate θ_{t+1} achieves a smaller value of the current loss.

From stationary optimization to online learning

In the Theorem 1.1 and the preceding Example 1.1, we assumed that, for every t, $\ell_t = f$, so the learner repeatedly optimizes the same fixed function, with the same minimizer $\theta^\star$. This is a stationary optimization problem and convergence toward $\theta^\star$ is therefore a meaningful goal. Since the purpose of online learning is to move to a genuinely sequential setting, even when the analytical form of the loss remains unchanged, the observation y_t varies with time, so that

$$\ell_t(\theta) = \ell(\theta; y_t).$$

The learner must choose θ_t using only past information, before y_t is revealed. Once y_t becomes available, it incurs $\ell_t(\theta_t) = \ell(\theta_t; y_t)$, computes

$$g_t \in \partial_\theta \ell(\theta_t; y_t),$$

and uses this information to construct θ_{t+1}.
Consequently, the loss function and its minimizer may change at every round, and the sequence (θ_t) **need not converge to a single fixed point**. The appropriate question is then no longer only whether the iterates converge, but whether their cumulative performance remains competitive with a clearly specified benchmark. This will motivate the definition of regret in the next sections.

1.4 From a scalar forecast to a parametric model

The scalar prediction $\widehat{y}_t = \theta_t$ is the simplest case. A richer forecaster uses a feature vector $x_t \in \mathbb{R}^d$, known when the prediction is made, and sets

$$\widehat{y}_t = x_t^\top \theta_t.$$

The scalar setting is recovered as the special case $x_t \equiv 1$, corresponding to a linear model with an intercept only, for which $\widehat{y}_t = x_t^\top \theta_t = \theta_t$. For supermarket demand, x_t could contain an intercept, recent sales, price, promotion indicators, day of the week, and holiday information. The online object is then the parameter vector θ_t, not merely the final prediction.

The running example will be developed in three complementary stages. The first two predict demand itself, and the third turns a demand forecast into an operational inventory decision:

1. Squared loss for online regression;
2. Squared loss with ridge regularization;
3. Pinball loss for an order quantity, with c_b measuring the unit cost of backorders and c_h the unit cost of holding excess inventory.

2 The online protocol and the first mathematical definitions

2.1 Decision set, loss sequence, and feedback

The forecasting story now needs a mathematical grammar that records who acts first and what becomes known afterward. The following protocol is deliberately abstract: once it is understood, predictions, probabilities, portfolios, and inventory levels all become instances of the same sequential decision problem.

Definition 2.1 (Online convex optimization protocol)**.** Let $\mathcal{K} \subseteq \mathbb{R}^d$ be a nonempty closed convex decision set *(meaning the collection of all admissible parameters or actions from which the learner is allowed to choose at each round)*. For rounds $t = 1, \dots, T$:

1. the learner chooses $\theta_t \in \mathcal{K}$ using only past information;
2. the environment reveals a convex loss $\ell_t : \mathcal{K} \to \mathbb{R}$;
3. the learner incurs $\ell_t(\theta_t)$ and receives sufficient feedback to compute a subgradient $g_t \in \partial \ell_t(\theta_t)$.

This is the *full-information* or *first-order feedback* setting.

Here, "online" means that decisions are made sequentially from the available information; we focus on first-order feedback, where each revealed loss provides a gradient or subgradient for the next update.

2.2 Convexity and subgradients

The learner must turn the loss revealed at the end of a round into useful information for the next round. Convexity is the structural assumption that makes this possible: a local slope then provides a valid global comparison, even when the loss has a corner and an ordinary derivative does not exist.

Definition 2.2 (Convex set and convex function)**.** A set $\mathcal{K} \subseteq \mathbb{R}^d$ is convex if $(1-\alpha)u+\alpha v \in \mathcal{K}$ for all $u, v \in \mathcal{K}$ and $\alpha \in [0, 1]$. A function $f : \mathcal{K} \to \mathbb{R}$ is convex if

$$f((1-\alpha)u + \alpha v) \le (1-\alpha)f(u) + \alpha f(v)$$

for the same choices of u, v, and α.

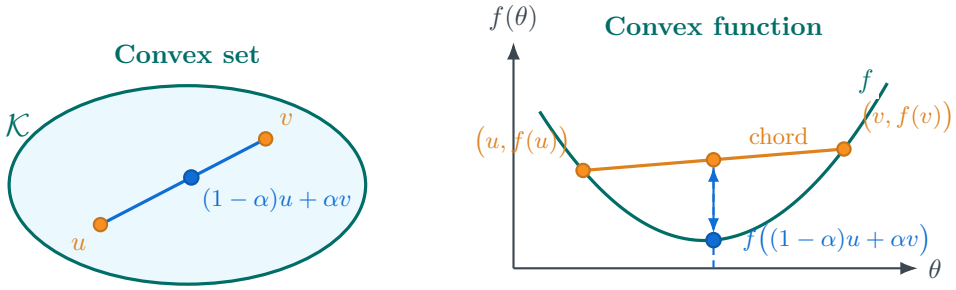


Figure 3: Geometric interpretation of convexity. On the left, every segment joining two points $u, v \in \mathcal{K}$ remains inside the convex set $\mathcal{K}$. On the right, the value of a convex function at a convex combination of u and v lies below the corresponding point on the chord joining $(u, f(u))$ and $(v, f(v))$.

Definition 2.3 (Subgradient)**.** A vector $g \in \mathbb{R}^d$ is a subgradient of a convex function f at $u \in \mathcal{K}$ if

$$\forall\, v \in \mathcal{K},\ \ f(v) \ge f(u) + \langle g, v - u \rangle\,.$$

The set of all such vectors is the subdifferential $\partial f(u)$. If f is differentiable at u, then $\partial f(u) = \{\nabla f(u)\}$.

Remark 2.1. Rearranging the subgradient inequality gives the fundamental online bound

$$f(u) - f(v) \le \langle g, u - v \rangle , \qquad g \in \partial f(u). \tag{3}$$

It allows us to replace nonlinear excess loss by a linear expression in a subgradient.

Example 2.1 (Absolute loss)**.** For $f(\theta) = |\theta - y|$, the derivative is -1 when $\theta < y$ and $+1$ when $\theta > y$. At $\theta = y$, the function is not differentiable, but every $g \in [-1, 1]$ is a valid subgradient. Online methods therefore do not require differentiability.

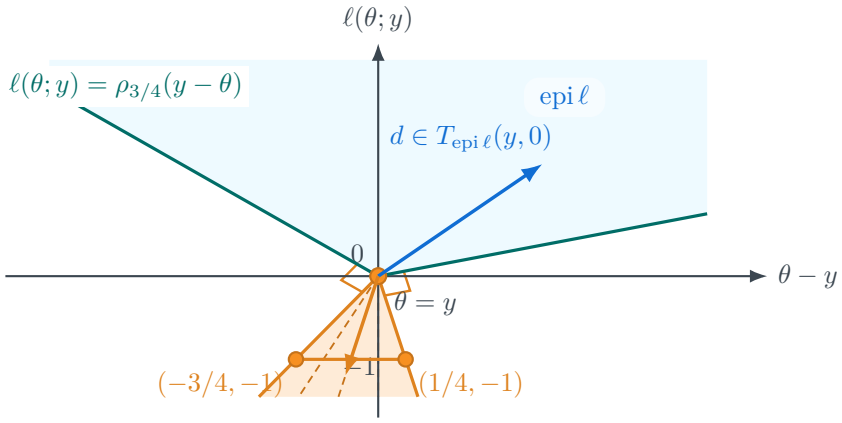
The pinball loss is the asymmetric version of this absolute loss. For a target level $\tau \in (0, 1)$, define

$$\rho_\tau(r) = \begin{cases} \tau r, & r \ge 0, \\ (\tau - 1)r, & r < 0. \end{cases}$$

For the prediction loss $\ell(\theta; y) = \rho_\tau(y - \theta)$, one may select as its gradient

$$g(\theta; y) \in \begin{cases} \{-\tau\}, & \theta < y, \\ [-\tau, 1 - \tau], & \theta = y, \\ \{1 - \tau\}, & \theta > y. \end{cases}$$

Thus a value below the observation and a value above it produce corrections of different magnitudes. This asymmetry will later encode the unequal costs of a stockout and excess inventory [6, 3].



$z \in N_{\text{epi}\,\ell}(y, 0) \implies \langle z, d \rangle \le 0$

$\partial_\theta \ell(y; y) = \{g : (g, -1) \in N_{\text{epi}\,\ell}(y, 0)\} = [-3/4, 1/4]$

Figure 4: Pinball loss for $\tau = 3/4$ and its normal cone at $\theta = y$. Every vector in the orange cone has a nonpositive inner product with every feasible direction in the epigraph tangent cone. The intersection of the normal cone with the line of ordinate -1 is $[-3/4, 1/4] \times \{-1\}$, which identifies the subdifferential $\partial_\theta \ell(y; y) = [-3/4, 1/4]$.

Optimality and the zero-subgradient condition. The subgradient inequality from Definition 2.3 gives more than a local approximation: it also provides a complete characterization of the minimizers of a convex function. At an optimum, the graph of the function admits a *horizontal supporting hyperplane*. Algebraically, this means that *the zero vector is a valid subgradient*.

Proposition 2.1 (First-order optimality for a convex function)**.** *Let $f : \mathcal{K} \to \mathbb{R}$ be convex, and interpret the subdifferential relative to $\mathcal{K}$ as in Definition 2.3. Then*

$$x^\star \in \arg\min_{x \in \mathcal{K}} f(x) \quad \Longleftrightarrow \quad 0 \in \partial f(x^\star).$$

If $x^\star$ is an interior point at which f is differentiable, this condition reduces to

$$\nabla f(x^\star) = 0.$$

Proof. Suppose first that $0 \in \partial f(x^\star)$. By the definition of a subgradient 2.3, for every $x \in \mathcal{K}$,

$$f(x) \geq f(x^\star) + \langle 0, x - x^\star \rangle = f(x^\star).$$

Thus no feasible point has a smaller value than $x^\star$, and therefore $x^\star$ is a global minimizer of f over $\mathcal{K}$.
Conversely, suppose that $x^\star$ minimizes f over $\mathcal{K}$. Then

$$f(x) \geq f(x^\star) = f(x^\star) + \langle 0, x - x^\star \rangle \qquad \text{for every } x \in \mathcal{K}.$$

Hence the zero vector satisfies the subgradient inequality at $x^\star$, so $0 \in \partial f(x^\star)$. □

The one-dimensional form. Let $f : I \to \mathbb{R}$ be convex on an interval $I \subseteq \mathbb{R}$, and let x belong to the interior of I. A convex function need not be differentiable, but its left and right derivatives exist and satisfy $f'_-(x) \leq f'_+(x)$. Moreover,

$$\partial f(x) = \left[f'_-(x), f'_+(x) \right]. \tag{4}$$

To understand (4), consider the slopes of the secant lines joining $(x, f(x))$ to the other points of the graph. Convexity implies that these slopes are ordered and that

$$f'_-(x) = \sup_{v < x} \frac{f(v) - f(x)}{v - x}, \qquad f'_+(x) = \inf_{v > x} \frac{f(v) - f(x)}{v - x}.$$

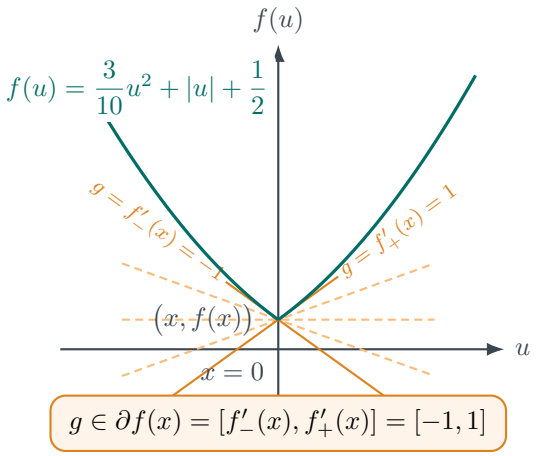


Figure 5: Supporting lines at a nondifferentiable point of a convex function.

Justification of (4). By Definition 2.3, a scalar $g \in \mathbb{R}$ belongs to $\partial f(x)$ if and only if

$$f(v) \geq f(x) + g \cdot (v - x) \qquad \text{for every } v \in I. \tag{$\star$}$$

- Suppose first that $g \in \partial f(x)$.
  - If $v > x$, then $v - x > 0$, so division by $v - x$ gives

$$(\star) \iff g \leq \frac{f(v) - f(x)}{v - x}.$$

    Since this inequality holds for every $v > x$, taking the infimum yields $g \leq f'_+(x)$.
  - If $v < x$, then $v - x < 0$, so the direction of the inequality is reversed:

$$(\star) \iff g \geq \frac{f(v) - f(x)}{v - x}.$$

    Taking the supremum over every $v < x$ gives $g \geq f'_-(x)$.

  Therefore, if $g \in \partial f(x)$, then $f'_-(x) \leq g \leq f'_+(x)$.
- Conversely, suppose that $f'_-(x) \leq g \leq f'_+(x)$.
  - For every $v > x$, the definition of $f'_+(x)$ as the infimum of the right-hand secant slopes gives

$$g \leq f'_+(x) \leq \frac{f(v) - f(x)}{v - x}.$$

    Since $v - x > 0$, this implies $f(v) \geq f(x) + g \cdot (v - x)$.
  - For every $v < x$, we similarly have

$$\frac{f(v) - f(x)}{v - x} \leq f'_-(x) \leq g.$$

    Multiplying by $v - x < 0$ reverses the inequality and again gives $f(v) \geq f(x) + g \cdot (v - x)$.

  Thus the subgradient inequality holds on both sides of x, and hence $g \in \partial f(x)$.

We conclude that

$$\partial f(x) = \{g \in \mathbb{R} : f'_-(x) \leq g \leq f'_+(x)\} = \left[f'_-(x), f'_+(x)\right].$$

□

Proposition 2.1 therefore becomes in Dimension 1 :

$$x^\star \in \arg\min_{x \in I} f(x) \quad \iff \quad f'_-(x^\star) \leq 0 \leq f'_+(x^\star). \tag{5}$$

Geometrically, the slope is nonpositive when approaching $x^\star$ from the left and "becomes" nonnegative when leaving it to the right. The function therefore stops decreasing and starts increasing at $x^\star$. If f is differentiable there, both one-sided derivatives coincide and (5) reduces to $f'(x^\star) = 0$.

For example, consider the absolute loss $f : x \mapsto |x - y|$ which its subdifferential is

$$\partial f(x) = \begin{cases} \{-1\}, & x < y, \\ [-1, 1], & x = y, \\ \{1\}, & x > y. \end{cases}$$

The zero subgradient belongs to $\partial f(x)$ only when $x = y$. Therefore,

$$\arg\min_{x \in \mathbb{R}} |x - y| = \{y\},$$

even though the function is not differentiable at its minimizer.

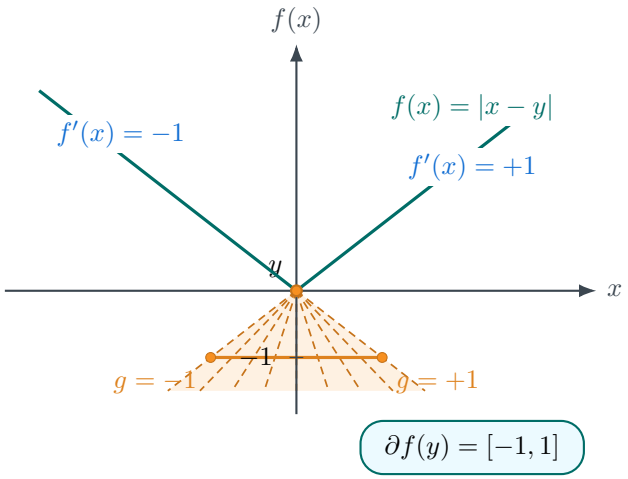


Figure 6: Absolute loss and its subdifferential at $x = y$.

2.3 Lipschitzness, smoothness, and strong convexity

Convexity tells us that first-order information is globally valid, but it does not tell us how large a step is safe or how fast regret can grow. The next three notions separate bounded slopes, slowly varying slopes, and genuine curvature; keeping these roles distinct prevents many incorrect rate claims.

Definition 2.4 (Three different regularity notions)**.** Let $f : \mathcal{K} \to \mathbb{R}$.

1. f is G-Lipschitz with respect to a norm $\|\cdot\|$ if $|f(u) - f(v)| \le G \|u - v\|$.
2. A differentiable f is β-smooth if $\|\nabla f(u) - \nabla f(v)\|_2 \le \beta \|u - v\|_2$.
3. f is α-strongly convex if, for every $g \in \partial f(u)$,

$$f(v) \ge f(u) + \langle g, v - u \rangle + \frac{\alpha}{2} \|v - u\|_2^2 .$$

These notions play different roles. Lipschitzness controls the magnitude of first-order changes; smoothness prevents the gradient from changing too abruptly; strong convexity supplies curvature in every direction. For differentiable convex losses on a convex set, a uniform bound $\|\nabla f(u)\|_2 \le G$ implies G-Lipschitzness on that set.

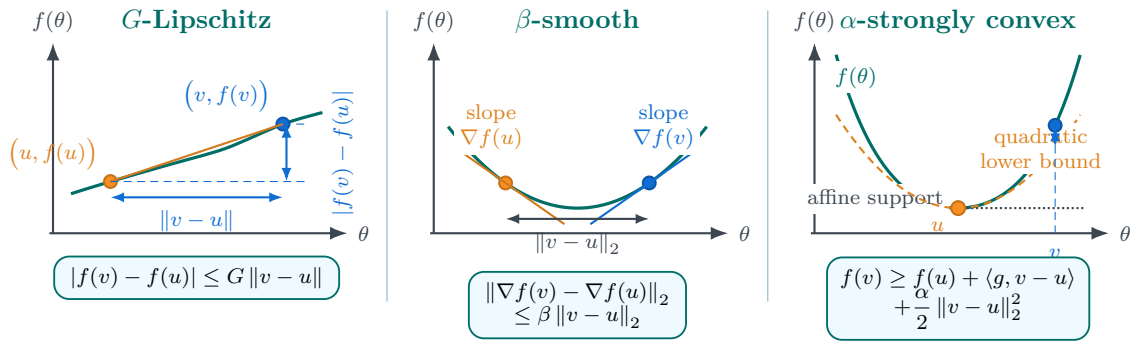


Figure 7: One-dimensional interpretation of the three regularity notions. Lipschitzness bounds changes in function values, smoothness bounds changes in gradients, and strong convexity places a quadratic lower bound beneath the function, thereby enforcing uniform curvature.

Remark 2.2. Consider the scalar squared loss

$$f(\theta) = \frac{1}{2}(\theta - y)^2.$$

Since $f'(\theta) = \theta - y$ and $f''(\theta) = 1$, the loss is 1-strongly convex. More explicitly, for every $u, v \in \mathbb{R}$,

$$f(v) = f(u) + f'(u)(v - u) + \frac{1}{2}(v - u)^2,$$

which is exactly the strong-convexity inequality with parameter $\alpha = 1$.
However, f is not globally Lipschitz on $\mathbb{R}$. Indeed,

$$\frac{|f(\theta) - f(y)|}{|\theta - y|} = \frac{\frac{1}{2}(\theta - y)^2}{|\theta - y|} = \frac{1}{2}|\theta - y| \longrightarrow \infty \qquad \text{as } |\theta| \to \infty.$$

Hence no finite constant G can satisfy $|f(u) - f(v)| \leq G|u - v|$ for all $u, v \in \mathbb{R}$. Equivalently, its derivative $f'(\theta) = \theta - y$ is unbounded.
Consequently, any later analysis requiring a uniform Lipschitz or gradient bound must either restrict the decisions and observations to bounded sets or prove separately that the iterates remain bounded.

2.4 Euclidean projection

A gradient step may leave the feasible set: a negative demand prediction or an order quantity above warehouse capacity is mathematically possible but operationally invalid. Projection is the correction that returns the tentative iterate to the nearest feasible decision.

Definition 2.5 (Projection onto a convex set)**.** For a nonempty closed convex set $\mathcal{K}$, the Euclidean projection of $z \in \mathbb{R}^d$ is

$$\Pi_{\mathcal{K}}(z) = \arg\min_{u \in \mathcal{K}} \|u - z\|_2 .$$

Proposition 2.2 (Projection does not increase distance to feasible points)**.** *For every $u \in \mathcal{K}$ and $z \in \mathbb{R}^d$,*

$$\|\Pi_{\mathcal{K}}(z) - u\|_2 \leq \|z - u\|_2 .$$

Proof. Let $z \in \mathbb{R}^d$ be fixed and set $p := \Pi_{\mathcal{K}}(z)$. By definition, p minimizes the function $\Phi : v \mapsto \frac{1}{2}\|v - z\|_2^2$ over all $v \in \mathcal{K}$.
Fix any $u \in \mathcal{K}$. Since $\mathcal{K}$ is convex, the segment

$$\gamma(s) := p + s(u - p), \qquad s \in [0, 1],$$

is contained in $\mathcal{K}$. Moreover, $\gamma(0) = p$, and increasing s moves from p toward u.
Consider the restriction of the objective Φ to this feasible segment γ:

$$\varphi(s) := \Phi(\gamma(s)) = \frac{1}{2}\|p + s(u - p) - z\|_2^2 .$$

Since p minimizes Φ over $\mathcal{K}$, the value $s = 0$ minimizes φ over $[0, 1]$. Its right derivative at 0 must therefore be nonnegative; otherwise, moving a sufficiently small distance from p toward u would

decrease the objective. Hence

$$0 \le \varphi'(0^+) = \frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{2}\|p + s(u-p) - z\|_2^2\right)\Big|_{s=0^+} = \langle p - z, u - p\rangle .$$

Expanding the squared norm gives

$$\|z-u\|_2^2 = \|z-p\|_2^2 + \|p-u\|_2^2 + 2\langle z-p, p-u\rangle .$$

Since $\langle z-p, p-u\rangle \ge 0$, we obtain

$$\|z-u\|_2^2 \ge \|p-u\|_2^2 .$$

Taking square roots proves the claim. □

Projection is what keeps a gradient update feasible. For $\mathcal{K} = [a,b]$, it is simply clipping: $\Pi_{[a,b]}(z) = \min\{b, \max\{a,z\}\}$.

Projection onto a Euclidean ball. A common decision set in online learning is the closed Euclidean ball $\mathcal{K} = \left\{\theta \in \mathbb{R}^d : \|\theta\|_2 \le R\right\}$. Here, $\mathcal{K}$ contains both the interior of the ball and its boundary $\theta : \|\theta\|_2 = R$. For any $z \in \mathbb{R}^d$, its Euclidean projection onto $\mathcal{K}$ is the feasible point closest to z:

$$\Pi_{\mathcal{K}}(z) = \arg\min_{u \in \mathcal{K}} \|u - z\|_2 .$$

For a ball centered at the origin, this projection has the explicit form

$$\Pi_{\mathcal{K}}(z) = \begin{cases} z, & \text{if } \|z\|_2 \le R, \\ \dfrac{R}{\|z\|_2} z, & \text{if } \|z\|_2 > R. \end{cases}$$

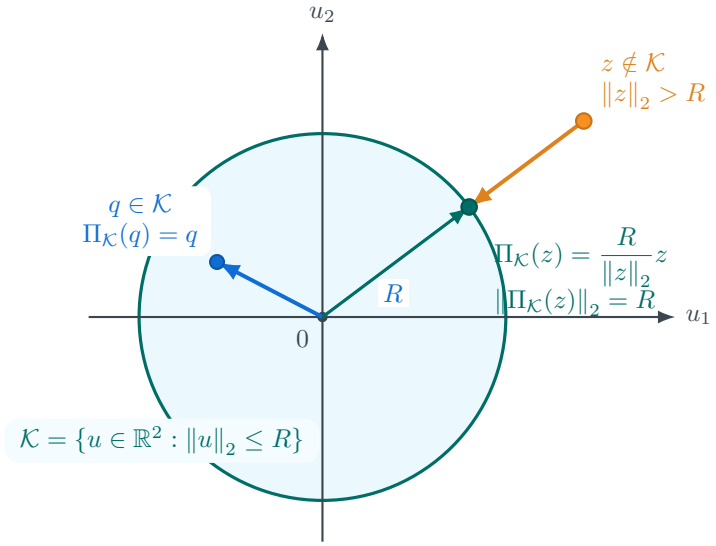


Figure 8: Euclidean projection onto a closed ℓ_2-ball. A point inside the ball remains unchanged, whereas an exterior point is moved radially to the closest point on the boundary.

Thus, a point already inside the ball remains unchanged. A point outside the ball is moved radially toward the origin until its norm becomes exactly R: its direction is preserved, while all its coordinates are multiplied by the same factor $R/\|z\|_2 < 1$. In projected OGD, one first computes the unconstrained update $z_{t+1} = \theta_t - \eta_t g_t$ and then sets $\theta_{t+1} = \Pi_{\mathcal{K}}(z_{t+1})$, thereby enforcing the constraint $\|\theta_{t+1}\|_2 \le R$.

2.5 Asymptotic notation and sublinear growth

Online learning studies quantities accumulated over a horizon of T rounds. Their total value may grow with T without making the algorithm ineffective. What matters is whether this growth is slower than the linear growth of the horizon itself. Asymptotic notation makes this comparison precise.

Definition 2.6 (Big-O, little-o, and sublinear growth)**.** Let $(a_T)_{T\ge1}$ be a real-valued sequence and let $(b_T)_{T\ge1}$ be positive for all sufficiently large T.

- We write $a_T = O(b_T)$ if there exist constants $C > 0$ and $T_0 \ge 1$ such that
$$|a_T| \le C \cdot b_T \qquad \text{for every } T \ge T_0.$$
- We write $a_T = o(b_T)$ if
$$\frac{|a_T|}{b_T} \xrightarrow[T\to\infty]{} 0 \qquad \text{as } T \to \infty.$$
- A sequence has *sublinear growth* when
$$a_T = o(T), \qquad \text{that is,} \qquad \frac{|a_T|}{T} \xrightarrow[T\to\infty]{} 0.$$

The notation $O(b_T)$ gives an asymptotic upper scale but does not claim that a_T is exactly proportional to b_T. For example,
$$a_T = O(\sqrt{T}) \quad \Longrightarrow \quad \frac{|a_T|}{T} \le \frac{C}{\sqrt{T}} \xrightarrow[T\to\infty]{} 0,$$
so every $O(\sqrt{T})$ sequence is sublinear. More generally, $O(1)$, $O(\log T)$, $O(T^\alpha)$ (with $\alpha \in [0, 1)$) are sublinear rates.

Later, this notion will be applied to cumulative excess loss: a sublinear upper bound means that the total discrepancy may grow with T, while its average excess per round becomes asymptotically negligible.

Sublinear does not mean bounded

A sublinear cumulative quantity may still diverge: both $\sqrt{T}$ and $\log T$ tend to infinity. Sublinearity means that this accumulation is slow relative to the number of rounds. In online learning, it ensures that the average price paid for not knowing the future **vanishes over time**.

3 How performance is measured: regret

3.1 The hindsight comparator

How should we evaluate a learner that must act before the next loss is known? Its cumulative loss alone is not sufficient: a large value may simply reflect an intrinsically difficult sequence. We therefore need a reference strategy facing exactly the same losses. A natural benchmark is the *best fixed decision that could have been selected after observing the entire sequence*.

Although this comparator benefits from hindsight, it is not allowed to adapt from one round to the next. The difference between the learner's cumulative loss and this retrospective benchmark measures the price paid for learning sequentially without knowing the future. This quantity is called the *regret*.

At the end of T rounds, define the cumulative loss of the learner as

$$L_T^{\text{alg}} = \sum_{t=1}^{T} \ell_t(\theta_t).$$

The standard comparator is the best *single fixed decision* in hindsight:

$$\theta_T^\star \in \arg\min_{u \in \mathcal{K}} \sum_{t=1}^{T} \ell_t(u).$$

This oracle sees the whole sequence before choosing u, but it must use the same u at every round.

Definition 3.1 (Static regret)**.** The regret relative to $u \in \mathcal{K}$ is

$$\text{Reg}_T(u) = \sum_{t=1}^{T} (\ell_t(\theta_t) - \ell_t(u)). \tag{6}$$

The static regret is

$$\text{Reg}_T = \sum_{t=1}^{T} \ell_t(\theta_t) - \min_{u \in \mathcal{K}} \sum_{t=1}^{T} \ell_t(u). \tag{7}$$

Definition 3.2 (No-reget)**.** An algorithm is said to be *no-regret* if

$$\frac{\text{Reg}_T}{T} \xrightarrow[T \to \infty]{} 0 \quad \Longleftrightarrow \quad \text{Reg}_T = o(T). \tag{8}$$

No-regret does not say that every decision is good or that the parameter converges. It says that the learner's *average excess loss* relative to the best fixed hindsight decision vanishes:

$$\frac{1}{T} \sum_{t=1}^{T} \ell_t(\theta_t) \leq \min_{u \in \mathcal{K}} \frac{1}{T} \sum_{t=1}^{T} \ell_t(u) + o(1).$$

A deliberately strong benchmark

The comparator has the informational advantage of hindsight, whereas the learner respects time. A sublinear regret bound proves that the price paid for this informational disadvantage becomes negligible per round.

3.2 First application to the running example

Before proving a general regret theorem, it is useful to compute the benchmark and the learner's losses by hand. This short example makes the chronology visible and shows exactly which two cumulative quantities are subtracted in the definition of regret.

Suppose $\mathcal{K} = [0, 20]$, $\widehat{y}_t = \theta_t$, and $\ell_t(\theta) = \frac{1}{2}(\theta - y_t)^2$. For the four demands $(y_1, y_2, y_3, y_4) = (12, 15, 11, 18)$, the best fixed (i.e., *constant*) prediction is the empirical mean

$$\theta_4^\star = \frac{1}{4}\sum_{i=1}^{4} y_i = 14.$$

Its cumulative loss is therefore given by

$$\sum_{i=1}^{4} \ell_i(\theta_4^\star) = \frac{1}{2}\sum_{i=1}^{4}(\theta_4^\star - y_i)^2 = \frac{1}{2}[(14-12)^2 + (14-15)^2 + (14-11)^2 + (14-18)^2] = 15.$$

Starting from $\theta_1 = 10$ and applying the update

$$\theta_{t+1} = \theta_t - 0.25(\theta_t - y_t)$$

gives the following fully chronological calculation. *Note that both the initialization θ_1 and the learning rate $\eta = 0.25$ are chosen for pedagogical convenience.*

t	θ_t chosen	y_t revealed	$g_t = \theta_t - y_t$	$\ell_t(\theta_t)$
1	10.000	12	-2.000	2.000
2	10.500	15	-4.500	10.125
3	11.625	11	0.625	0.195
4	11.469	18	-6.531	21.329

Using the exact values before rounding, the learner's cumulative online loss is

$$L_4^{\text{alg}} = \sum_{t=1}^{4} \ell_t(\theta_t) = \frac{1}{2}\sum_{t=1}^{4}(\theta_t - y_t)^2 \approx 33.65.$$

The best fixed prediction in hindsight is $\theta_4^\star = 14$, whose cumulative loss is

$$L_4^\star = \min_{\theta \in \mathcal{K}} \sum_{t=1}^{4} \ell_t(\theta) = \sum_{t=1}^{4} \ell_t(\theta_4^\star) = 15.$$

Therefore, the four-round static regret is

$$\text{Reg}_4 = L_4^{\text{alg}} - L_4^\star \approx 18.65.$$

This large short-horizon value is **not** a contradiction: the no-regret property is asymptotic, whereas the present calculation contains only four rounds and starts from a prediction substantially below the hindsight optimum.

If the same decisions were order quantities evaluated by pinball loss, the hindsight comparator would no longer be the empirical mean. It would be an empirical quantile whose level is fixed by the relative backorder and holding costs. Static regret would then measure excess total inventory cost relative to the best single order quantity chosen after all demands were known.

3.3 Static regret is not tracking error

If demand changes structurally, the best fixed predictor may itself be a poor operational benchmark. A more demanding alternative is the dynamic regret

$$\mathrm{Reg}_T^{\mathrm{dyn}}(u_{1:T}) = \sum_{t=1}^{T} (\ell_t(\theta_t) - \ell_t(u_t)),$$

where the comparator u_t may vary over time. Without restricting this variation, the worst-case dynamic regret may grow linearly with T, so no sublinear guarantee is possible. A standard restriction is to control the comparator's path length $\sum_{t=2}^{T} \|u_t - u_{t-1}\|$ as in [9]. Static regret asks "*Did the learner compete with the best fixed decision?*", whereas dynamic regret asks "*Did the learner track a moving comparator whose total movement was limited?*"

The rest of this chapter uses static regret because it provides the cleanest introduction to online performance guarantees.

4 Several fundamental online algorithms

Defining regret tells us what good online performance means, but not how to achieve it. The central algorithmic difficulty is to balance two competing requirements: the learner must *react to newly observed losses*, **yet** it must *remain stable enough* not to overreact to a single round. The methods introduced below can be understood as successive answers to this problem. Despite their differences, all these methods use past feedback to construct the next decision without accessing future information.

Throughout this section, g_t denotes the first-order information obtained from the loss revealed at round t. More precisely, it is a subgradient of ℓ_t evaluated at the current decision θ_t: $g_t \in \partial \ell_t(\theta_t)$. This means that, for every $u \in \mathcal{K}$, $\ell_t(u) \geq \ell_t(\theta_t) + \langle g_t, u - \theta_t \rangle$. Thus, g_t defines an affine lower approximation of the current loss and indicates how the parameter should be modified after observing round t. When ℓ_t is differentiable, the subgradient is unique and simply equals the usual gradient: $g_t = \nabla \ell_t(\theta_t)$. When ℓ_t is nonsmooth, several valid subgradients may exist, and the algorithm may select any one of them.

4.1 Follow the Leader: fit all past losses

The most immediate strategy is to *behave tomorrow as the best fixed decision would have behaved on all observations seen so far*. This rule is appealing because it uses the regret comparator itself as an update, but the example after the algorithm shows that this retrospective optimality does not always lead to stable online behavior.

Follow the Leader (FTL)

Choose an arbitrary $\theta_1 \in \mathcal{K}$. After observing ℓ_t, set

$$\theta_{t+1} \in \arg\min_{u \in \mathcal{K}} \sum_{s=1}^{t} \ell_s(u). \tag{9}$$

Thus the next decision is the empirical minimizer of all losses observed so far.

FTL is natural but can be unstable.

Example 4.1. Let $\mathcal{K} = [-1, 1]$ and

$$\ell_t(\theta) = g_t\theta, \qquad g_t = \begin{cases} 1, & t \text{ odd}, \\ -1, & t \text{ even}. \end{cases}$$

At the beginning of round t, FTL chooses

$$\theta_t \in \operatorname*{arg\,min}_{\theta\in[-1,1]} \left(\sum_{s=1}^{t-1} g_s\right)\theta.$$

If t is odd, then the past gradients cancel:

$$\sum_{s=1}^{t-1} g_s = 0.$$

The cumulative past loss is therefore equal to zero for every $\theta \in [-1, 1]$, so every decision is a minimizer. To make the algorithm completely specified, suppose that it chooses $\theta_t = +1$ whenever this nonuniqueness occurs. This convention is called a *tie-breaking rule*.

If t is even, then

$$\sum_{s=1}^{t-1} g_s = 1,$$

so FTL minimizes θ over $[-1, 1]$ and uniquely chooses $\theta_t = -1$. Consequently,

$$\theta_t = \begin{cases} +1, & t \text{ odd}, \\ -1, & t \text{ even}. \end{cases}$$

Since g_t has the same sign as θ_t, the learner incurs

$$\ell_t(\theta_t) = g_t\theta_t = 1$$

at every round. For even T, the gradients cancel over the whole sequence, and every fixed decision $u \in [-1, 1]$ has cumulative loss

$$\sum_{t=1}^{T} \ell_t(u) = u\sum_{t=1}^{T} g_t = 0.$$

Thus the learner suffers cumulative loss T, whereas the best fixed decision suffers loss 0, giving $\mathrm{Reg}_T = T$.

This example shows that repeatedly minimizing past losses can be highly sensitive to one additional observation.

For pinball loss, the same rule has a concrete interpretation: FTL orders at an empirical quantile of the demands observed so far. That interpretation is useful, but the quantile can still jump when a new observation changes the relevant order statistic; regularization or a gradient step offers a smoother response.

4.2 Follow the Regularized Leader: stabilize the leader

The previous example reveals the main weakness of FTL: when several decisions have nearly identical cumulative losses, one additional observation may abruptly change the empirical minimizer. *Follow the Regularized Leader* addresses this instability by adding a fixed, curved function R to the cumulative objective. This regularizer discourages extreme decisions and makes the next leader less sensitive to small changes in the observed loss sequence.

Follow the Regularized Leader (FTRL)

Let $R : \mathcal{K} \to \mathbb{R}$ be a strongly convex regularizer and let $\eta > 0$. Initialize

$$\theta_1 \in \arg\min_{u \in \mathcal{K}} R(u).$$

After observing the loss ℓ_t, update

$$\theta_{t+1} \in \arg\min_{u \in \mathcal{K}} \left\{ \sum_{s=1}^{t} \ell_s(u) + \frac{1}{\eta} R(u) \right\}. \tag{10}$$

The parameter η controls the trade-off between adaptation and regularization. A small η gives a large weight $1/\eta$ to R, producing more conservative decisions. A large η weakens the regularization and makes FTRL behave more like the original FTL rule.

A particularly important version of FTRL replaces each convex loss by its *first-order approximation* at the decision θ_s:

$$\ell_s(u) \approx \ell_s(\theta_s) + \langle g_s, u - \theta_s \rangle, \qquad g_s \in \partial \ell_s(\theta_s).$$

Since the terms $\ell_s(\theta_s) - \langle g_s, \theta_s \rangle$ do not depend on u, they do not affect the minimizer. FTRL applied to these linearized losses therefore becomes

$$\theta_{t+1} \in \arg\min_{u \in \mathcal{K}} \left\{ \left\langle \sum_{s=1}^{t} g_s, u \right\rangle + \frac{1}{\eta} R(u) \right\}.$$

For the quadratic regularizer $R(u) = \frac{1}{2}\|u\|_2^2$ and the unconstrained domain $\mathcal{K} = \mathbb{R}^d$, define the objective function

$$J(u) := \left\langle \sum_{s=1}^{t} g_s, u \right\rangle + \frac{1}{2\eta} \|u\|_2^2.$$

The vectors $g_1, \ldots, g_t$ are fixed when differentiating with respect to u. Therefore,

$$\nabla_u J(u) = \sum_{s=1}^{t} g_s + \frac{1}{\eta} u.$$

Moreover, since $\nabla_u^2 J(u) = \frac{1}{\eta} I_d \succ 0$, (because $\eta > 0$), J is therefore strongly convex and admits a unique global minimizer.

Consequently,

$$\theta_{t+1} = \arg\min_{u \in \mathbb{R}^d} J(u)$$

is characterized by the first-order optimality condition $\nabla_u J(\theta_{t+1}) = 0$. This yields:

$$\nabla_u J(\theta_{t+1}) = 0 \iff \theta_{t+1} = -\eta \sum_{s=1}^{t} g_s.$$

FTRL accumulates all past first-order information, while the quadratic curvature prevents the minimization problem from selecting arbitrarily extreme leaders. More generally, the choice of R determines the geometry and stability properties of the algorithm.

4.3 Online Gradient Descent: one local step

Re-solving a full optimization problem after every observation may be unnecessary or too expensive. OGD instead asks a smaller question: using the newly revealed slope, what single feasible step should be taken before the next round?

Projected Online Gradient Descent (OGD)

Initialization. Choose $\theta_1 \in \mathcal{K}$ and positive learning rates $(\eta_t)_{t\geq 1}$. For each round $t = 1, \ldots, T$:

1. **History.** The learner has access to
$$\mathcal{H}_{t-1} = \{(\theta_s, \ell_s(\theta_s), g_s) : 1 \leq s \leq t-1\},$$
2. **Decision.** Play $\theta_t \in \mathcal{K}$ (through previous Step 4. **Update** for $t \geq 2$) before observing ℓ_t.
3. **Feedback.** Observe $\ell_t(\theta_t)$ and sufficient information to select $g_t \in \partial \ell_t(\theta_t)$.
4. **Update.** Set
$$\theta_{t+1} = \Pi_{\mathcal{K}}(\theta_t - \eta_t g_t).$$
Store θ_{t+1}: it becomes the decision played at round $t+1$.

Although the complete history $\mathcal{H}_{t-1}$ is available in principle, standard OGD does not need to process all previous iterates again. Their influence has already been summarized recursively in the current iterate θ_t. Consequently, once g_t has been computed, the update requires only θ_t, g_t, and η_t; the vectors $\theta_1, \ldots, \theta_{t-1}$ need not be stored. If the horizon ends at round T, the final iterate θ_{T+1} is not played, but it remains useful for the analysis or for continuing the data stream.

The update has a physical interpretation. The vector $-g_t$ points in a direction that locally decreases the current loss. The step size η_t determines how strongly one observation can change the model. Projection enforces feasibility. OGD uses constant memory apart from the current parameter and is typically linear in dimension once the gradient is known.

Keeping our previous main example. Suppose that the learner must produce a scalar demand prediction in the feasible interval $\mathcal{K} = [0, 20]$. After demand y_t is revealed, the prediction is evaluated with the squared loss

$$\ell_t(\theta) = \frac{1}{2}(\theta - y_t)^2, \qquad g_t = \nabla \ell_t(\theta_t) = \theta_t - y_t.$$

Choose $\theta_1 = 18$ and the constant learning rate $\eta_t = 0.25$. These values are selected only to make the calculation transparent.

First round. The learner first plays $\theta_1 = 18$. Demand is then revealed to be $y_1 = 30$, so the incurred loss is $\ell_1(\theta_1) = \frac{1}{2}(18 - 30)^2 = 72$. The gradient is $g_1 = \theta_1 - y_1 = 18 - 30 = -12$. Because the prediction was too low, the gradient is negative and the descent direction $-g_1$ is positive. The unconstrained update would be

$$\widetilde{\theta}_2 = \theta_1 - \eta_1 g_1 = 18 - 0.25(-12) = 21.$$

However, $21 \notin [0, 20]$. Projection therefore enforces feasibility:

$$\theta_2 = \Pi_{[0,20]}(\widetilde{\theta}_2) = \Pi_{[0,20]}(21) = 20.$$

Second round. The learner plays $\theta_2 = 20$ before observing the new demand. Suppose that $y_2 = 14$ is then revealed. The gradient becomes $g_2 = \theta_2 - y_2 = 20 - 14 = 6$. The prediction is now too high, so the positive gradient produces a downward correction:

$$\widetilde{\theta}_3 = \theta_2 - \eta_2 g_2 = 20 - 0.25 \times 6 = 18.5.$$

Since $18.5 \in [0, 20]$, projection has no effect:

$$\theta_3 = \Pi_{[0,20]}(18.5) = 18.5.$$

This example separates the three components of projected OGD. The gradient determines the direction of correction, the learning rate determines its magnitude, and projection modifies the tentative update only when it violates the decision constraints. Most importantly, θ_t is always selected before y_t is observed.

4.4 Online Mirror Descent: choose a geometry

Euclidean distance is not always appropriate. Probability vectors live on a simplex, positive variables have asymmetric geometry, and sparse or high-dimensional problems may be better described by non-Euclidean norms.

Definition 4.1 (Bregman divergence)**.** Let ψ be differentiable and strictly convex. Its Bregman divergence is

$$D_\psi(u, v) = \psi(u) - \psi(v) - \langle \nabla\psi(v), u - v \rangle .$$

It is nonnegative, but it is generally neither symmetric nor a metric.

This definition is a direct consequence of the first-order inequality for convex functions. Indeed, convexity of ψ gives

$$\psi(u) \geq \psi(v) + \langle \nabla\psi(v), u - v \rangle ,$$

and therefore

$$D_\psi(u, v) = \psi(u) - \Big[\psi(v) + \langle \nabla\psi(v), u - v \rangle\Big] \geq 0.$$

Thus, $D_\psi(u, v)$ is precisely the gap between $\psi(u)$ and the first-order affine approximation of ψ constructed at v. It measures the curvature ignored by this linearization, relative to the reference point v. In Online Mirror Descent, it replaces Euclidean squared distance and penalizes updates according to the geometry encoded by ψ.

Online Mirror Descent (OMD)

Given $g_t \in \partial\ell_t(\theta_t)$, update

$$\theta_{t+1} \in \arg\min_{u \in \mathcal{K}} \{\eta_t \langle g_t, u \rangle + D_\psi(u, \theta_t)\} .$$

With $\psi(u) = \frac{1}{2}\|u\|_2^2$, $D_\psi(u,v) = \frac{1}{2}\|u-v\|_2^2$ and OMD reduces to OGD. On the probability simplex, negative entropy $\psi(u) = \sum_j u_j \log u_j$ yields multiplicative updates,

$$\theta_{t+1,j} = \frac{\theta_{t,j}\exp(-\eta_t g_{t,j})}{\sum_k \theta_{t,k}\exp(-\eta_t g_{t,k})}.$$

This is the exponentiated-gradient or Hedge geometry used for learning with expert advice [4, 7].

4.5 AdaGrad: let the data scale each coordinate

A single learning rate can be poorly calibrated when coordinates have different scales. A diagonal AdaGrad [2] update uses

$$H_{t,j} = \sqrt{\varepsilon + \sum_{s=1}^{t} g_{s,j}^2}, \qquad \theta_{t+1,j} = \theta_{t,j} - \eta\frac{g_{t,j}}{H_{t,j}},$$

followed by projection if needed. Coordinates that have accumulated large gradients receive smaller future steps. AdaGrad can be derived as adaptive mirror descent or adaptive FTRL. This illustrates a general lesson: many named algorithms differ mainly in the geometry, regularizer, and learning-rate schedule they use.

Summary Table of Methods

Algorithm	Update principle	Main design choice
FTL	Minimize all past losses	No explicit stabilization
FTRL	Minimize past losses plus a regularizer	Stability through R
OGD	Take one projected gradient step	Euclidean geometry and η_t
OMD	Take one step in Bregman geometry	Mirror map ψ
AdaGrad	Rescale coordinates using past gradients	Data-adaptive geometry

5 The core regret analysis of Follow the Leader and FTRL

The algorithms introduced in the previous section are easy to state, but their updates alone do not explain why they should perform well. The central question is whether a learner that sees the losses only sequentially can compete with the best fixed decision selected after the entire sequence has been observed.

For Follow the Leader, the answer depends on *stability*: does the minimizer of the cumulative past loss move only slightly when one new loss is added? For Follow the Regularized Leader, stability is deliberately created by adding curvature to the cumulative objective. This section makes these two mechanisms precise.

> **Remark 5.1** (What "no regret" means)**.** A no-regret guarantee does not generally mean that $\mathrm{Reg}_T \to 0$. The cumulative regret may increase with T. The relevant requirement is sublinear growth $\mathrm{Reg}_T = o(T)$. Thus, the additional loss paid for making decisions without knowing the future becomes negligible *per round*.

5.1 The leader that is allowed to see the current loss

For $t \geq 1$, define the **cumulative loss**

$$L_t(u) := \sum_{s=1}^{t} \ell_s(u), \qquad L_0(u) := 0.$$

Recall that the FTL decisions satisfy (9), i.e., $\theta_{t+1} \in \arg\min_{u\in\mathcal{K}} L_t(u)$.

The index shift is important. The decision θ_t is feasible online because it depends only on $\ell_1, \dots, \ell_{t-1}$. By contrast, θ_{t+1} already uses the current loss ℓ_t. It cannot be played at round t, but it provides a useful fictitious benchmark for the analysis.

Proposition 5.1 (Be-the-Leader inequality)**.** *Assume that the cumulative minimizers θ_{t+1} exist. Then, for every $u \in \mathcal{K}$,*

$$\sum_{t=1}^{T} \ell_t(\theta_{t+1}) \le \sum_{t=1}^{T} \ell_t(u). \tag{11}$$

In particular,

$$\sum_{t=1}^{T} \ell_t(\theta_{t+1}) \le \min_{u \in \mathcal{K}} \sum_{t=1}^{T} \ell_t(u).$$

Proof. We argue by induction on T. For $T = 1$, the result follows from

$$\theta_2 \in \arg\min_{u \in \mathcal{K}} \ell_1(u).$$

Suppose that the result holds up to round $T-1$. Applying the induction hypothesis with the comparator θ_T gives

$$\sum_{t=1}^{T-1} \ell_t(\theta_{t+1}) \le \sum_{t=1}^{T-1} \ell_t(\theta_T) = L_{T-1}(\theta_T).$$

Because θ_T minimizes L_{T-1},

$$L_{T-1}(\theta_T) \le L_{T-1}(\theta_{T+1}).$$

Consequently,

$$\sum_{t=1}^{T} \ell_t(\theta_{t+1}) \le L_{T-1}(\theta_{T+1}) + \ell_T(\theta_{T+1}) = L_T(\theta_{T+1}) \le L_T(u),$$

where the last inequality follows because θ_{T+1} minimizes L_T. □

The proposition says that a sequence allowed to use the current loss before acting performs at least as well as every fixed comparator. The actual learner cannot use this sequence directly; its regret therefore comes from the delay between θ_t and θ_{t+1} [7].

5.2 FTL regret is controlled by stability

Proposition 5.2 (FTL stability bound)**.** *For every comparator $u \in \mathcal{K}$, Follow the Leader satisfies*

$$\mathrm{Reg}_T(u) \le \sum_{t=1}^{T} \left(\ell_t(\theta_t) - \ell_t(\theta_{t+1})\right). \tag{12}$$

Proof. Add and subtract the losses of the one-step-ahead leaders:

$$\mathrm{Reg}_T(u) = \sum_{t=1}^{T} (\ell_t(\theta_t) - \ell_t(u)) = \sum_{t=1}^{T} (\ell_t(\theta_t) - \ell_t(\theta_{t+1})) + \sum_{t=1}^{T} (\ell_t(\theta_{t+1}) - \ell_t(u)).$$

The second sum is nonpositive by Proposition 5.1, which proves the result. □

This result identifies the exact difficulty faced by FTL. The quantity

$$\ell_t(\theta_t) - \ell_t(\theta_{t+1})$$

measures the price paid at round t because the learner used the previous leader rather than the leader obtained after seeing the current loss.

If each loss is G-Lipschitz, then

$$\ell_t(\theta_t) - \ell_t(\theta_{t+1}) \le G \|\theta_t - \theta_{t+1}\|_2 ,$$

and therefore

$$\mathrm{Reg}_T(u) \le G \sum_{t=1}^{T} \|\theta_t - \theta_{t+1}\|_2 . \tag{13}$$

Thus, FTL is no-regret whenever the total movement of its leaders is sublinear.

The alternating linear-loss example from the previous section violates precisely this requirement: the leader repeatedly jumps between the two ends of the interval, so its total movement grows linearly with T.

The weakness of unregularized FTL

FTL does not fail because minimizing the past is an unreasonable idea. It fails when the minimizer of the past is too sensitive to one additional loss. A useful regret guarantee must therefore control how much the leader can move between consecutive rounds.

5.3 Strong convexity stabilizes Follow the Leader

Strong convexity makes the cumulative minimizer resistant to small changes in the objective. If every loss contributes curvature, then the cumulative loss L_t becomes increasingly curved as t grows. This forces consecutive leaders to become progressively more stable.

The following elementary inequality will be used repeatedly:

$$ar - \frac{b}{2} r^2 \le \frac{a^2}{2b}, \qquad a \ge 0, \quad b > 0, \quad r \ge 0. \tag{14}$$

It follows by completing the square.

Theorem 5.1 (Logarithmic regret of FTL under strong convexity)**.** *Assume that every loss ℓ_t is α-strongly convex on $\mathcal{K}$, with the same constant $\alpha > 0$. Suppose that the selected subgradients satisfy*

$$g_t \in \partial \ell_t(\theta_t), \qquad \|g_t\|_2 \le G.$$

Then Follow the Leader satisfies, for every $u \in \mathcal{K}$,

$$\mathrm{Reg}_T(u) \le \frac{G^2}{2\alpha} \sum_{t=1}^{T} \frac{1}{t} \le \frac{G^2}{2\alpha}(1 + \log T). \tag{15}$$

Proof. We first use the following refined leader decomposition:

$$\mathrm{Reg}_T(u) = \sum_{t=1}^{T} (L_t(\theta_t) - L_t(\theta_{t+1})) + L_T(\theta_{T+1}) - L_T(u). \tag{16}$$

Indeed, substituting $L_t(\theta_t) = L_{t-1}(\theta_t) + \ell_t(\theta_t)$ makes the intermediate cumulative-loss terms telescope.

Since θ_{T+1} is defined as FTL (9), it minimizes L_T:

$$L_T(\theta_{T+1}) - L_T(u) \leq 0.$$

It remains to control $L_t(\theta_t) - L_t(\theta_{t+1})$.
Since each loss ℓ_s is α-strongly convex, their strong-convexity parameters add when the losses are summed. Therefore, the cumulative loss $L_{t-1} = \sum_{s=1}^{t-1} \ell_s$ is $(t-1)\alpha$-strongly convex.
Moreover, θ_t minimizes L_{t-1} over $\mathcal{K}$. The quadratic-growth property of a strongly convex function at its minimizer therefore gives

$$L_{t-1}(\theta_{t+1}) \geq L_{t-1}(\theta_t) + \frac{(t-1)\alpha}{2} \|\theta_{t+1} - \theta_t\|_2^2 .$$

With strong convexity of ℓ_s giving

$$\ell_s(\theta_{s+1}) \geq \ell_s(\theta_s) + \langle g_s, \theta_{s+1} - \theta_s \rangle + \frac{\alpha}{2} \|\theta_{s+1} - \theta_s\|_2^2 ,$$

adding these inequalities from 1 to t yields

$$L_t(\theta_{t+1}) \geq L_t(\theta_t) + \langle g_t, \theta_{t+1} - \theta_t \rangle + \frac{t\alpha}{2} \|\theta_{t+1} - \theta_t\|_2^2 .$$

Therefore,

$$\begin{aligned}
L_t(\theta_t) - L_t(\theta_{t+1}) &\leq \langle g_t, \theta_t - \theta_{t+1} \rangle - \frac{t\alpha}{2} \|\theta_t - \theta_{t+1}\|_2^2 && \text{(from the preceding inequality)} \\
&\leq \|g_t\|_2 \|\theta_t - \theta_{t+1}\|_2 - \frac{t\alpha}{2} \|\theta_t - \theta_{t+1}\|_2^2 && \text{(Cauchy–Schwarz inequality)} \\
&\leq \frac{\|g_t\|_2^2}{2t\alpha} && \text{(completing the square from (14))} \\
&\leq \frac{G^2}{2t\alpha} && \text{(since } \|g_t\|_2 \leq G\text{)}.
\end{aligned}$$

Substituting this bound into Eq. (16) gives

$$\mathrm{Reg}_T(u) \leq \frac{G^2}{2\alpha} \sum_{t=1}^{T} \frac{1}{t}.$$

Finally,

$$\sum_{t=1}^{T} \frac{1}{t} \leq 1 + \int_1^T \frac{\mathrm{d}x}{x} = 1 + \log T,$$

hence the result (15). □

The important mechanism is the factor $t\alpha$. Each new strongly convex loss adds curvature, so the cumulative objective becomes more stable over time. The price paid for following the previous leader is consequently of order $1/t$, whose cumulative sum is logarithmic.

5.4 FTRL creates the stability that linear losses lack

The preceding theorem is reassuring, but it relies on curvature already being present in every loss. Linear losses provide no such curvature, and their cumulative minimizer may jump abruptly. FTRL addresses this problem by adding a strongly convex regularizer.

Consider linearized FTRL:

$$\theta_t \in \arg\min_{v\in\mathcal{K}} \left\{ \sum_{s=1}^{t-1} \langle g_s, v\rangle + \frac{1}{\eta} R(v) \right\}, \tag{17}$$

where $g_s \in \partial \ell_s(\theta_s)$ and R is strongly convex.

The regularizer has two related effects. It keeps the optimization problem curved even when the linearized losses are flat, and it limits the movement caused by adding one new gradient.

Theorem 5.2 (Regret of linearized FTRL)**.** *Let $\mathcal{K} \subseteq \mathbb{R}^d$ be nonempty, closed, and convex. Suppose that $R : \mathcal{K} \to \mathbb{R}$ is α-strongly convex, with $\alpha > 0$, and initialize*

$$\theta_1 \in \arg\min_{v\in\mathcal{K}} R(v).$$

Let every loss ℓ_t be convex and select $g_t \in \partial \ell_t(\theta_t)$. Then the update (17) satisfies, for every $u \in \mathcal{K}$,

$$\mathrm{Reg}_T(u) \le \frac{R(u) - R(\theta_1)}{\eta} + \frac{\eta}{2\alpha} \sum_{t=1}^{T} \|g_t\|_2^2 . \tag{18}$$

Proof. Convexity gives $\ell_t(\theta_t) - \ell_t(u) \le \langle g_t, \theta_t - u\rangle /$ It is therefore sufficient to control the cumulative linearized regret.
Define

$$F_t(v) := \frac{1}{\eta} R(v) + \sum_{s=1}^{t-1} \langle g_s, v\rangle .$$

By construction, $\theta_t \in \arg\min_{v\in\mathcal{K}} F_t(v)$. A direct telescoping calculation gives

$$\sum_{t=1}^{T} \langle g_t, \theta_t - u\rangle = \frac{R(u) - R(\theta_1)}{\eta} + \sum_{t=1}^{T} \left(F_{t+1}(\theta_t) - F_{t+1}(\theta_{t+1})\right) + F_{T+1}(\theta_{T+1}) - F_{T+1}(u). \tag{19}$$

Since θ_{T+1} minimizes F_{T+1}, the last line is nonpositive.
Because R is α-strongly convex, F_t is $\frac{\alpha}{\eta}$-strongly convex. Since θ_t minimizes F_t,

$$F_t(\theta_{t+1}) \ge F_t(\theta_t) + \frac{\alpha}{2\eta} \|\theta_{t+1} - \theta_t\|_2^2 .$$

Using

$$F_{t+1}(v) = F_t(v) + \langle g_t, v\rangle ,$$

we obtain

$$\begin{aligned} F_{t+1}(\theta_t) - F_{t+1}(\theta_{t+1}) &= F_t(\theta_t) - F_t(\theta_{t+1}) + \langle g_t, \theta_t - \theta_{t+1}\rangle && \text{(definition of } F_{t+1}) \\ &\le \langle g_t, \theta_t - \theta_{t+1}\rangle - \frac{\alpha}{2\eta}\,\|\theta_t - \theta_{t+1}\|_2^2 && \text{(quadratic growth of } F_t \text{ at its minimizer } \theta_t) \\ &\le \|g_t\|_2\,\|\theta_t - \theta_{t+1}\|_2 - \frac{\alpha}{2\eta}\,\|\theta_t - \theta_{t+1}\|_2^2 && \text{(Cauchy–Schwarz inequality)} \\ &\le \frac{\eta}{2\alpha}\,\|g_t\|_2^2 && \text{(completing the square from (14)).} \end{aligned}$$

Substituting into Eq. (19) proves the result. □

The two terms in Eq. (18) describe the learning-rate trade-off:

$$\underbrace{\frac{R(u) - R(\theta_1)}{\eta}}_{\text{cost of competing with } u} + \underbrace{\frac{\eta}{2\alpha}\sum_{t=1}^{T}\|g_t\|_2^2}_{\text{accumulated adaptation cost}}.$$

A small η gives the regularizer a large weight and makes the learner too conservative. A large η weakens the regularizer and allows each new gradient to move the leader more strongly.

Proposition 5.3 (Corollary: Square-root regret of FTRL)**.** *Assume, in addition, that $\|g_t\|_2 \le G$ and define, for $R : \mathcal{K} \to \mathbb{R}$ a α-strongly convex function ($\alpha > 0$),*

$$D_R := \max_{u\in\mathcal{K}}\left(R(u) - \min_{v\in\mathcal{K}} R(v)\right).$$

Then

$$\mathrm{Reg}_T \le \frac{D_R}{\eta} + \frac{\eta G^2 T}{2\alpha}.$$

If the horizon is known, the choice $\eta = \frac{\sqrt{2\alpha D_R}}{G\sqrt{T}}$ gives

$$\mathrm{Reg}_T \le G\sqrt{\frac{2D_R T}{\alpha}}. \tag{20}$$

Consequently,

$$\frac{\mathrm{Reg}_T}{T} \le G\sqrt{\frac{2D_R}{\alpha T}} \xrightarrow[T\to\infty]{} 0.$$

Proof. The first inequality follows from Theorem 5.2 and $\sum_{t=1}^{T}\|g_t\|_2^2 \le G^2 T$. The two terms are equal for the stated choice of η, which gives Eq. (20). □

If the horizon T is unknown, a time-varying regularization weight or a doubling schedule gives the same $O(\sqrt{T})$ order [4, 7].

Example 5.1 (Euclidean FTRL)**.** Let

$$R(u) = \frac{1}{2}\|u\|_2^2, \qquad \mathcal{K} = \{u \in \mathbb{R}^d : \|u\|_2 \le B\}.$$

The regularizer is 1-strongly convex and $D_R = B^2/2$. Linearized FTRL becomes

$$\theta_{t+1} \in \arg\min_{u\in\mathcal{K}} \left\{ \left\langle \sum_{s=1}^{t} g_s, u \right\rangle + \frac{1}{2\eta}\|u\|_2^2 \right\}.$$

With $\eta = B/(G\sqrt{T})$, Corollary 5.3 gives

$$\mathrm{Reg}_T \le BG\sqrt{T}.$$

Thus the quadratic regularizer converts the potentially unstable leader into a no-regret algorithm, even when the original losses provide no curvature.

5.5 What the two analyses teach us

FTL and FTRL obtain stability in two different ways.

For FTL with strongly convex losses, curvature is supplied by the data: after t rounds, the cumulative loss has curvature of order $t\alpha$. The leaders therefore become increasingly stable, producing the harmonic sum

$$\sum_{t=1}^{T} \frac{1}{t}$$

and logarithmic regret.

For FTRL with general convex losses, curvature is supplied by the regularizer. The regularization strength and the learning rate must be balanced, leading to the general $O(\sqrt{T})$ rate. More refined versions of FTRL allow time-varying, proximal, or nonsmooth regularizers and connect FTRL with mirror descent and sparse online learning [7, 8].

Why vanishing average regret matters

FTL under strong convexity and FTRL under general convexity satisfy, respectively,

$$\frac{\mathrm{Reg}_T}{T} = O\left(\frac{\log T}{T}\right) \qquad \text{and} \qquad \frac{\mathrm{Reg}_T}{T} = O\left(\frac{1}{\sqrt{T}}\right).$$

In both cases, the average price of making decisions without knowing the future vanishes. Over a long horizon, the learner's average loss is therefore asymptotically no larger than that of the best fixed decision selected in hindsight by an oracle that has observed the complete loss sequence.
This does not imply that the parameters θ_t converge, that every individual prediction becomes optimal, or that the learner tracks a moving oracle. It is a cumulative guarantee relative to one fixed comparator.

The common principle

FTL succeeds when the observed losses themselves stabilize the leader. FTRL succeeds by adding the curvature required for stability. In both analyses, no regret follows from controlling how much the preferred decision can change when one new piece of information is added.

6 The core regret analysis of Online Gradient Descent

6.1 The one-step inequality

To prove the regret bound, we first study a single update. For any fixed comparator u, we compare the distance from θ_t to u before the update with the distance from θ_{t+1} to u afterward. A decrease in this distance will help us bound the learner's excess loss at round t.

Standing assumptions. Assume throughout this section that $\mathcal{K}$ has Euclidean diameter at most D, meaning that

$$\|u - v\|_2 \le D \quad \text{for all } u, v \in \mathcal{K},$$

and that the selected subgradients satisfy $\|g_t\|_2 \le G$ for a constant $G \ge 0$.

Proposition 6.1 (Potential decrease in one OGD step)**.** *For every comparator $u \in \mathcal{K}$, projected OGD satisfies*

$$\langle g_t, \theta_t - u \rangle \le \frac{\|\theta_t - u\|_2^2 - \|\theta_{t+1} - u\|_2^2}{2\eta_t} + \frac{\eta_t}{2} \|g_t\|_2^2 . \tag{21}$$

Proof. By the distance-reducing property of projection,

$$\|\theta_{t+1} - u\|_2^2 = \|\Pi_{\mathcal{K}}(\theta_t - \eta_t g_t) - u\|_2^2 \le \|\theta_t - \eta_t g_t - u\|_2^2 .$$

Expanding the right-hand side gives

$$\|\theta_{t+1} - u\|_2^2 \le \|\theta_t - u\|_2^2 - 2\eta_t \langle g_t, \theta_t - u \rangle + {\eta_t}^2 \|g_t\|_2^2 .$$

Rearranging and dividing by $2\eta_t$ proves the inequality. □

The difference $\ell_t(\theta_t) - \ell_t(u)$ is the learner's excess loss at round t relative to u: it measures how much more the learner loses than it would have lost by playing u on the same round; if it is negative, the learner performs better.

Convexity bounds this difference by $\langle g_t, \theta_t - u \rangle$. Proposition 6.1 then controls this quantity using the decrease of the *potential* $\|\theta_t - u\|_2^2$ and the gradient cost $\frac{\eta_t}{2}\|g_t\|_2^2$. When the inequalities are summed with a constant learning rate, the intermediate potential terms cancel by telescoping, which turns the one-step bounds into a cumulative regret bound.

6.2 The standard square-root regret bound

The one-step inequality (21) matters because its distance terms cancel across time. Convexity links current excess loss to the remaining inner product, so summing the local comparisons produces a global guarantee.

Theorem 6.1 (OGD regret for convex Lipschitz losses)**.** *Suppose every ℓ_t is convex, $\|g_t\|_2 \le G$, and $\mathcal{K}$ has diameter at most D. With a constant step size $\eta > 0$, projected OGD satisfies, for every $u \in \mathcal{K}$,*

$$\mathrm{Reg}_T(u) \le \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}.$$

If the horizon T is known, choosing $\eta = D/(G\sqrt{T})$ yields

$$\mathrm{Reg}_T \le DG\sqrt{T}, \qquad \frac{\mathrm{Reg}_T}{T} \le \frac{DG}{\sqrt{T}} \xrightarrow[T\to\infty]{} 0. \tag{22}$$

Proof. Convexity and (3) imply $\ell_t(\theta_t) - \ell_t(u) \le \langle g_t, \theta_t - u \rangle$. Recall from Eq. (6) that

$$\mathrm{Reg}_T(u) = \sum_{t=1}^{T} (\ell_t(\theta_t) - \ell_t(u)).$$

Apply Proposition 6.1 with $\eta_t = \eta$, use $\|g_t\|_2 \le G$, and sum over t:

$$\mathrm{Reg}_T(u) \le \frac{1}{2\eta} \sum_{t=1}^{T} \left(\|\theta_t - u\|_2^2 - \|\theta_{t+1} - u\|_2^2 \right) + \frac{\eta G^2 T}{2}.$$

The potential terms telescope. Moreover, since $\theta_1, u \in \mathcal{K}$ where $\mathcal{K}$ has diameter at most D, we have $\|\theta_1 - u\|_2 \le D$. Using also $\|\theta_{T+1} - u\|_2^2 \ge 0$, we obtain

$$\mathrm{Reg}_T(u) \le \frac{\|\theta_1 - u\|_2^2 - \|\theta_{T+1} - u\|_2^2}{2\eta} + \frac{\eta G^2 T}{2} \le \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}.$$

The two terms are balanced by $\eta = D/(G\sqrt{T})$, giving the result. □

The bound exposes the learning-rate trade-off. A small η makes the initial-distance term $D^2/(2\eta)$ large: the learner adapts too slowly. A large η makes the accumulated-gradient term $\eta G^2 T/2$ large: the learner overreacts.

Balancing the two gives the characteristic $\sqrt{T}$ rate. This rate is worst-case optimal for general convex Lipschitz losses up to constants [4, 7]. If T is unknown, one may take $\eta_t = D/(G\sqrt{t})$ and obtain a bound of the same order, for example $\mathrm{Reg}_T \le \frac{3}{2} DG\sqrt{T}$ in Hazan's analysis [4].

6.3 Strong convexity improves the rate

The $O(\sqrt{T})$ regret bound applies to general convex losses, including losses that may be almost flat and therefore provide little information about how far the current decision is from a good one.

If every loss ℓ_t is α-strongly convex with the same constant $\alpha > 0$, then its graph lies at least $\frac{\alpha}{2}\|v - u\|_2^2$ above each first-order linearization (see Figure 7). Thus, moving away from the comparator produces an additional cost that grows quadratically with the distance. This quadratic term gives tighter control of $\|\theta_t - u\|_2^2$ and allows the learning rate to decrease over time, ultimately leading to a sharper regret bound.

Theorem 6.2 (Logarithmic regret under strong convexity)**.** *Assume each loss is α-strongly convex and $\|g_t\|_2 \le G$. Projected OGD with $\eta_t = 1/(\alpha t)$ satisfies*

$$\mathrm{Reg}_T \le \frac{G^2}{2\alpha}(1 + \log T). \tag{23}$$

Proof. Strong convexity gives, for every $u \in \mathcal{K}$,

$$\ell_t(\theta_t) - \ell_t(u) \le \langle g_t, \theta_t - u \rangle - \frac{\alpha}{2}\|\theta_t - u\|_2^2.$$

Combining this with the one-step inequality (21) from Proposition 6.1 and $\|g_t\|_2 \le G$ for all $t \ge 1$ yields

$$\ell_t(\theta_t) - \ell_t(u) \le \frac{\|\theta_t - u\|_2^2 - \|\theta_{t+1} - u\|_2^2}{2\eta_t} - \frac{\alpha}{2}\|\theta_t - u\|_2^2 + \frac{\eta_t G^2}{2}. \tag{$*$}$$

After telescoping, some terms will vanish. To make the cancellation explicit, let's define $A_t :=$

$\|\theta_t - u\|_2^2$. Summing the preceding inequality $(*)$ from $t = 1$ to T gives

$$\mathrm{Reg}_T(u) \le \sum_{t=1}^{T} \left(\frac{1}{2\eta_t} - \frac{\alpha}{2} \right) A_t - \sum_{t=1}^{T} \frac{1}{2\eta_t} A_{t+1} + \frac{G^2}{2} \sum_{t=1}^{T} \eta_t.$$

Reindexing the second sum with $s = t + 1$ gives

$$\sum_{t=1}^{T} \frac{1}{2\eta_t} A_{t+1} = \sum_{s=2}^{T+1} \frac{1}{2\eta_{s-1}} A_s.$$

Collecting the coefficients of the potential terms yields

$$\mathrm{Reg}_T(u) \le \underbrace{\left(\frac{1}{2\eta_1} - \frac{\alpha}{2} \right)}_{=0} A_1 + \sum_{t=2}^{T} \underbrace{\left(\frac{1}{2\eta_t} - \frac{1}{2\eta_{t-1}} - \frac{\alpha}{2} \right)}_{=0} A_t - \frac{1}{2\eta_T} A_{T+1} + \frac{G^2}{2} \sum_{t=1}^{T} \eta_t.$$

Since $A_{T+1} \ge 0$, then $-\frac{1}{2\eta_T} A_{T+1} \le 0$. We obtain

$$\mathrm{Reg}_T(u) \le \frac{G^2}{2} \sum_{t=1}^{T} \eta_t = \frac{G^2}{2\alpha} \sum_{t=1}^{T} \frac{1}{t} \le \frac{G^2}{2\alpha} (1 + \log T),$$

where the last inequality follows from the sum–integral comparison lemma[a]

$$\sum_{t=1}^{T} \frac{1}{t} \le 1 + \int_1^T \frac{\mathrm{d}x}{x} = 1 + \log T.$$

Taking u to be the best fixed comparator proves the theorem. □

[a] See full proof of Theorem 1.3 in [7] for great pedagogical details.

Strong convexity changes the geometry of the problem: moving away from the comparator produces a quadratic penalty, which cancels the distance terms created by the time-varying learning rate.

Why these regret bounds matter

Equations (22) and (23) provide *finite-time certificates*: under the stated assumptions, they control the additional cumulative loss caused by making each decision before observing the corresponding loss.

- ▷ The first bound shows that this price grows only as $O(\sqrt{T})$. Hence, although the cumulative regret may increase with time, its average contribution **vanishes**:

$$\frac{\mathrm{Reg}_T}{T} = O\left(\frac{1}{\sqrt{T}} \right).$$

  The online learner therefore becomes *asymptotically as effective, on average*, as the **best fixed decision** chosen with full hindsight.

- ▷ These distribution-free bounds enable principled comparisons between algorithms while showing how the diameter D of the decision set, the subgradient bound G, and the strong-convexity parameter α govern the problem's difficulty.

However, small static regret does not by itself imply that θ_t converges, nor that the learner tracks a time-varying optimum; it guarantees cumulative performance relative to the best *fixed* comparator.

6.4 Back to the supermarket: what the two regret bounds imply

Return to the supermarket example. Each morning, the learner predicts a scalar demand $\theta_t \in \mathcal{K} = [0, 20]$, and the realized demand satisfies $y_t \in [0, 20]$. Consider the squared loss

$$\ell_t(\theta) = \frac{1}{2}(\theta - y_t)^2.$$

The best fixed prediction in hindsight is the empirical mean

$$\bar{y}_T = \frac{1}{T}\sum_{t=1}^{T} y_t,$$

which belongs to $[0, 20]$. The corresponding regret is

$$\mathrm{Reg}_T^{\mathrm{sq}} = \sum_{t=1}^{T} \frac{1}{2}(\theta_t - y_t)^2 - \sum_{t=1}^{T} \frac{1}{2}(\bar{y}_T - y_t)^2.$$

It measures the additional cumulative squared error caused by making predictions sequentially, relative to the best constant demand forecast chosen after observing the entire sequence.

Using only convexity. The decision set has diameter $D = 20$, and $\nabla \ell_t(\theta) = \theta - y_t$. Since both θ and y_t belong to $[0, 20]$, $|\nabla \ell_t(\theta)| = |\theta - y_t| \le 20$, so one may take $G = 20$. If we use only convexity and bounded gradients, Equation (22), with the constant learning rate $\eta = \frac{D}{G\sqrt{T}} = 1/\sqrt{T}$, gives

$$\mathrm{Reg}_T^{\mathrm{sq}} \le DG\sqrt{T} = 400\sqrt{T}. \tag{24}$$

Consequently,

$$\frac{\mathrm{Reg}_T^{\mathrm{sq}}}{T} \le \frac{400}{\sqrt{T}}.$$

This guarantee is valid, but it ignores an important property of the squared loss.

Using strong convexity. For every round t, $\nabla^2 \ell_t(\theta) = 1$. Thus each squared loss is 1-strongly convex, so $\alpha = 1$. Projected OGD with $\eta_t = 1/t$ can therefore use Equation (23), which gives

$$\mathrm{Reg}_T^{\mathrm{sq}} \le 200(1 + \log T), \qquad \frac{\mathrm{Reg}_T^{\mathrm{sq}}}{T} \le \frac{200(1 + \log T)}{T}. \tag{25}$$

The loss function and the demand sequence are exactly the same as before. The improvement comes only from using the additional curvature of the squared loss in the analysis.

For illustration, the two average-regret guarantees are

Horizon T	General convex bound	Strongly convex bound
100	$\mathrm{Reg}_T^{\mathrm{sq}}/T \le 40$	$\mathrm{Reg}_T^{\mathrm{sq}}/T \le 11.21$
10 000	$\mathrm{Reg}_T^{\mathrm{sq}}/T \le 4$	$\mathrm{Reg}_T^{\mathrm{sq}}/T \le 0.204$

These numbers are worst-case certificates, not predictions of the regret that will be observed in practice. Such bounds can be loose for a particular demand sequence. Their main message is the difference in asymptotic behavior:

$$\frac{1}{\sqrt{T}} \qquad \text{versus} \qquad \frac{\log T}{T}.$$

The second quantity decreases much faster. Hence, when strong convexity is genuinely available, exploiting it gives a substantially sharper long-horizon guarantee.

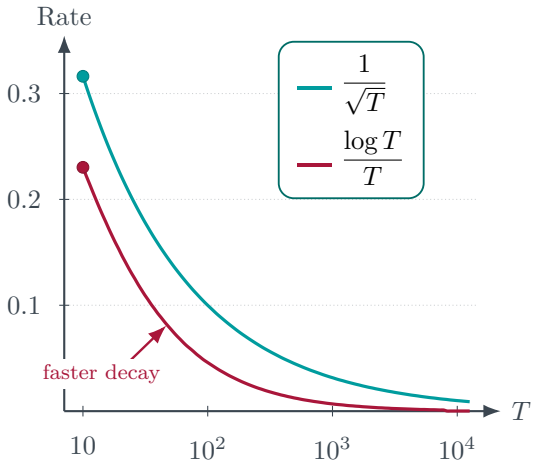


Figure 9: Comparison of the average-regret rates $1/\sqrt{T}$ and $\log(T)/T$ on a logarithmic scale. Both converge to zero, but the strongly convex rate $\log(T)/T$ decreases substantially faster for large horizons.

What if the store uses another loss? The applicable regret bound depends on the geometry of the loss function. We consider three important cases.

Absolute loss. Suppose the store evaluates its forecasts using $\ell_t(\theta) = |\theta - y_t|$, for $\theta, y_t \in [0, 20]$. This loss is convex and its subgradients satisfy $|g_t| \leq 1.$ for all $t \geq 1$. However, it is not strongly convex because it is linear on each side of y_t. Since the decision set has diameter $D = 20$, Theorem 6.1–or equivalently Equation (22)– applies with $G = 1$, gives

$$\mathrm{Reg}_T^{\mathrm{abs}} \leq 20\sqrt{T}.$$

The logarithmic bound from Theorem 6.2 cannot be applied because the absolute loss has no uniform quadratic curvature.

Pinball inventory loss. Consider the pinball inventory loss (studied later in Section 9)

$$\ell_t(q) = c_h(q - y_t)_+ + c_b(y_t - q)_+, \qquad q \in [0, Q_{\max}].$$

Its subgradients satisfy for all $t \geq 1$, $|g_t| \leq G_{\mathrm{inv}}$, with $G_{\mathrm{inv}} := \max\{c_b, c_h\}$. The feasible interval has diameter $D = Q_{\max}$. Since the loss is convex, Theorem 6.1 gives

$$\mathrm{Reg}_T^{\mathrm{inv}} \leq Q_{\max} G_{\mathrm{inv}} \sqrt{T} = Q_{\max} \max\{c_b, c_h\} \sqrt{T}.$$

Again, Theorem 6.2 does not apply because the pinball loss is piecewise linear and therefore not strongly convex.

Squared loss with ridge regularization. Suppose now that the store uses features $x_t \in \mathbb{R}^d$ and the regularized loss, for $\lambda > 0$,

$$\widetilde{\ell}_t(\theta) = \frac{1}{2}(x_t^\top \theta - y_t)^2 + \frac{\lambda}{2}\|\theta\|_2^2.$$

Its Hessian is given by $\nabla^2 \widetilde{\ell}_t(\theta) = x_t x_t^\top + \lambda I_d \succeq \lambda I_d$. Thus every $\widetilde{\ell}_t$ is λ-strongly convex, even when $x_t x_t^\top$ is rank deficient. Assume that $\|\theta\|_2 \le B$, $\|x_t\|_2 \le R$, $|y_t| \le Y$. Then, we will show in Subsection 8.3 that

$$\left\|\nabla \widetilde{\ell}_t(\theta)\right\|_2 \le (RB + Y)R + \lambda B =: G_\lambda.$$

Theorem 6.2, applied with $\alpha = \lambda$ and $\eta_t = 1/(\lambda \cdot t)$, therefore gives

$$\mathrm{Reg}_T^{\mathrm{ridge}} \le \frac{G_\lambda^2}{2\lambda}(1 + \log T).$$

This is a logarithmic guarantee for regret measured with the regularized losses $\widetilde{\ell}_t$.

The loss determines both the objective and the rate

The regret rate depends on both the algorithm and the geometry of the loss. Convex losses with bounded subgradients, such as absolute and pinball losses, yield the general $O(\sqrt{T})$ guarantee, whereas uniform curvature–provided directly by the loss or added through ridge regularization–makes an $O(\log T)$ guarantee available.

A faster theoretical rate is not the only criterion for choosing a loss. Although the pinball loss is not strongly convex, it directly represents the asymmetric costs of shortage and excess inventory, targets the economically relevant demand quantile, and produces simple bounded updates. It may therefore be more appropriate operationally than a strongly convex but poorly aligned loss.

Finally, regret bounds should only be compared when they concern the same objective: logarithmic ridge regret controls cumulative regularized loss, not unregularized prediction error alone.

7 Detailed application I: online regression with squared loss

7.1 From a scalar forecast to online linear regression

Section 6.4 studied a scalar demand forecast. We now allow the supermarket to use information available before day t, such as price, promotions, recent sales, or calendar effects.

Let $x_t \in \mathbb{R}^d$ be the feature vector observed before the prediction. The learner chooses $\theta_t \in \mathcal{K}$ and predicts

$$\widehat{y}_t = x_t^\top \theta_t.$$

After demand y_t is revealed, the squared loss is

$$\ell_t(\theta) = \frac{1}{2}(x_t^\top \theta - y_t)^2.$$

Writing $e_t = x_t^\top \theta_t - y_t$ for the prediction residual, the gradient and Hessian are

$$\nabla \ell_t(\theta) = (x_t^\top \theta - y_t)x_t, \qquad \nabla^2 \ell_t(\theta) = x_t x_t^\top. \tag{26}$$

The Hessian is positive semidefinite since, for every $v \in \mathbb{R}^d$, $v^\top x_t x_t^\top v = (x_t^\top v)^2 \ge 0$. Hence the loss is convex.

In dimension $d > 1$, however, $x_t x_t^\top$ has rank at most one. A single observation therefore supplies curvature only in the direction x_t and provides no information in directions orthogonal to x_t. Consequently, an individual squared loss is generally not strongly convex in the full parameter space.

7.2 The OGD update

Applying projected OGD gives

$$\theta_{t+1} = \Pi_{\mathcal{K}}\Big(\theta_t - \eta_t(x_t^\top \theta_t - y_t)x_t\Big).$$

The correction is proportional to both the residual and the feature vector. If $e_t > 0$, the prediction is too high and the algorithm moves in the direction $-x_t$. If $e_t < 0$, the prediction is too low and the algorithm moves in the direction x_t. A larger prediction error produces a larger update.

Example 7.1 (One multivariate update)**.** Let

$$x_t = [1,\, 2]^\top, \quad \theta_t = [3,\, 1]^\top, \quad y_t = 7, \quad \eta_t = 0.1.$$

The prediction and residual are $\widehat{y}_t = x_t^\top \theta_t = 5$, and $e_t = 5 - 7 = -2$. Therefore,

$$g_t = e_t x_t = [-2,\, -4]^\top.$$

If projection is inactive, the next parameter is

$$\theta_{t+1} = \theta_t - \eta_t g_t = \begin{pmatrix} 3.2 \\ 1.4 \end{pmatrix}.$$

Both coefficients increase because, for this positive feature vector, increasing either coordinate raises the predicted demand.

7.3 A bounded-gradient regret guarantee

To apply Theorem 6.1, assume $\mathcal{K} = \Big\{\theta \in \mathbb{R}^d : \|\theta\|_2 \le B\Big\}$, $\|x_t\|_2 \le R$, and $|y_t| \le Y$. For every $\theta \in \mathcal{K}$, the prediction residual satisfies

$$\begin{aligned} |x_t^\top \theta - y_t| &\le |x_t^\top \theta| + |y_t| && \text{(triangle inequality)} \\ &\le \|x_t\|_2 \, \|\theta\|_2 + |y_t| && \text{(Cauchy–Schwarz inequality)} \\ &\le RB + Y && \text{(boundedness assumptions).} \end{aligned}$$

Recall that Eq. (26) gives $\nabla \ell_t(\theta) = (x_t^\top \theta - y_t)x_t,$, leading to

$$\begin{aligned} \|\nabla \ell_t(\theta)\|_2 &= \Big\|(x_t^\top \theta - y_t)x_t\Big\|_2 \\ &= |x_t^\top \theta - y_t| \, \|x_t\|_2 && \text{(absolute homogeneity)} \\ &\le \underbrace{(RB + Y)R.}_{=:G_{\text{sq}}} && \text{(previous residual bound)} \end{aligned}$$

The Euclidean ball $\mathcal{K}$ has diameter $D = 2B$. Therefore, Theorem 6.1 with $\eta = 2B/(G_{\text{sq}}\sqrt{T})$, gives

$$\text{Reg}_T^{\text{sq}} \le 2B(RB + Y)R\sqrt{T}.$$

Thus the average excess squared loss is bounded above by

$$\frac{\mathrm{Reg}_T^{\mathrm{sq}}}{T} \le \frac{2B(RB+Y)R}{\sqrt{T}},$$

which vanishes as the horizon grows.

Remark 7.1. Unlike the scalar intercept-only case, we cannot directly invoke Theorem 6.2: each matrix $x_t x_t^\top$ may be singular. The cumulative matrix $\sum_{t=1}^T x_t x_t^\top$ can eventually become positive definite if the features explore enough directions, but this does not imply that every individual loss is uniformly strongly convex, as required by that theorem.

7.4 What the comparator represents

For squared online regression, the best fixed comparator is

$$u_T^\star \in \arg\min_{u\in\mathcal{K}} \frac{1}{2}\sum_{t=1}^T (x_t^\top u - y_t)^2.$$

It is the best constrained least-squares coefficient computed after the entire sequence has been observed. Hence

$$\mathrm{Reg}_T^{\mathrm{sq}} = \sum_{t=1}^T \frac{1}{2}(x_t^\top \theta_t - y_t)^2 - \frac{1}{2}\sum_{t=1}^T (x_t^\top u_T^\star - y_t)^2$$

compares the sequential coefficients $\theta_1, \dots, \theta_T$ with one fixed regression vector.

The comparator knows the complete dataset, but it cannot change its coefficient from one day to the next. Low static regret therefore does not guarantee that the learner *tracks a changing relationship between features and demand*. Such a question would require a dynamic comparator and a dynamic-regret analysis.

What squared loss gives us

Squared loss is useful when large prediction errors should be penalized more heavily than small ones. Its gradient produces a correction whose magnitude is proportional to the observed residual, and its hindsight comparator has the familiar interpretation of a least-squares fit.
In the scalar case, squared loss is strongly convex and gives a logarithmic regret guarantee. In multivariate regression, one observation usually identifies only one feature direction, so the robust conclusion is the $O(\sqrt{T})$ bound. This reveals that the rate depends not only on the formula of the loss, but also on the geometry of the features.

8 Detailed application II: squared loss with ridge

8.1 Ridge controls poorly observed directions

The previous application showed that the multivariate squared loss may be flat in directions orthogonal to x_t. This can make coefficients unstable when features are redundant, highly correlated, or rarely observed. Ridge regularization adds curvature in every direction.

For $\lambda > 0$, define

$$\widetilde{\ell}_t(\theta) = \frac{1}{2}(x_t^\top \theta - y_t)^2 + \frac{\lambda}{2}\|\theta\|_2^2.$$

Its gradient and Hessian are respectivly expressed as

$$\nabla \widetilde{\ell}_t(\theta) = (x_t^\top \theta - y_t)x_t + \lambda\theta, \quad \nabla^2 \widetilde{\ell}_t(\theta) = x_t x_t^\top + \lambda I_d \succeq \lambda I_d. \tag{27}$$

Every regularized loss is λ-strongly convex, even when $x_t x_t^\top$ is rank deficient.

8.2 The ridge OGD update

Substituting the regularized gradient into projected OGD gives

$$\theta_{t+1} = \Pi_{\mathcal{K}}\Big((1-\eta_t\lambda)\theta_t - \eta_t(x_t^\top\theta_t - y_t)x_t\Big).$$

This update combines two distinct mechanisms:

$$\underbrace{(1-\eta_t\lambda)\theta_t}_{\text{ridge shrinkage}} \qquad \text{and} \qquad \underbrace{-\eta_t(x_t^\top\theta_t - y_t)x_t}_{\text{data correction}}.$$

When $0 < \eta_t\lambda < 1$, the first term contracts every coefficient toward zero. The second term responds to the current prediction error only in the observed feature direction.

Example 8.1 (Shrinking an unobserved direction)**.** Let

$$x_t = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The data-fitting term contains no information about the second coordinate. Without ridge, that coordinate is unchanged by the gradient update. With ridge, and when projection is inactive,

$$\theta_{t+1} = (1-\eta_t\lambda)\theta_t.$$

Thus a coefficient unsupported by the current data is gradually moved toward zero instead of remaining at an arbitrary previous value.

8.3 The logarithmic guarantee and its exact objective

The ridge update is not only a heuristic for shrinking coefficients. By adding the Hessian term λI_d, ridge supplies the uniform curvature that is missing from multivariate squared loss. We can therefore convert the local update into a finite-time guarantee on the learner's cumulative performance.

The relevant benchmark is the best *fixed regularized coefficient* that could have been selected after observing the entire sequence. The objective is not to prove that every iterate θ_t converges, but to show that the average additional regularized loss caused by learning sequentially becomes negligible.

Proposition 8.1 (Uniform curvature and bounded gradients)**.** *Suppose that*

$$\mathcal{K} = \{\theta \in \mathbb{R}^d : \|\theta\|_2 \le B\}, \qquad \|x_t\|_2 \le R, \qquad |y_t| \le Y,$$

and let $\lambda > 0$. Then, for every round t:

1. *$\widetilde{\ell}_t$ is λ-strongly convex on $\mathcal{K}$;*
2. *its gradient satisfies*

$$\left\|\nabla\widetilde{\ell}_t(\theta)\right\|_2 \le (RB+Y)R + \lambda B =: G_\lambda \qquad \text{for every } \theta \in \mathcal{K}.$$

Proof. From Eq. (27), $\nabla^2\widetilde{\ell}_t(\theta) = x_t x_t^\top + \lambda I_d$. For every direction $h \in \mathbb{R}^d$,

$$h^\top\nabla^2\widetilde{\ell}_t(\theta)h = h^\top x_t x_t^\top h + \lambda h^\top h = (x_t^\top h)^2 + \lambda\|h\|_2^2 \ge \lambda\|h\|_2^2.$$

Hence $\nabla^2\widetilde{\ell}_t(\theta) \succeq \lambda I_d$, which proves that $\widetilde{\ell}_t$ is λ-strongly convex.

For the gradient bound, the triangle inequality gives

$$\left\|\nabla\widetilde{\ell}_t(\theta)\right\|_2 = \left\|(x_t^\top\theta - y_t)x_t + \lambda\theta\right\|_2 \le |x_t^\top\theta - y_t|\,\|x_t\|_2 + \lambda\,\|\theta\|_2\,.$$

Moreover,

$$\begin{aligned} |x_t^\top\theta - y_t| &\le |x_t^\top\theta| + |y_t| && \text{(triangle inequality)}\\ &\le \|x_t\|_2\,\|\theta\|_2 + |y_t| && \text{(Cauchy–Schwarz inequality)}\\ &\le RB + Y && \text{(boundedness assumptions).}\end{aligned}$$

Consequently,

$$\left\|\nabla\widetilde{\ell}_t(\theta)\right\|_2 \le (RB+Y)R + \lambda B = G_\lambda.$$

□

The proposition verifies the two assumptions required by the strongly convex OGD theorem: every loss has the same minimum amount of curvature, and no individual gradient can be arbitrarily large on the feasible set.

Theorem 8.1 (Logarithmic regularized regret for ridge OGD)**.** *Under the assumptions of Proposition 8.1, projected ridge OGD with $\eta_t = 1/(\lambda t)$ satisfies, for every fixed comparator $u \in \mathcal{K}$,*

$$\sum_{t=1}^{T}\left(\widetilde{\ell}_t(\theta_t) - \widetilde{\ell}_t(u)\right) \le \frac{G_\lambda^2}{2\lambda}(1+\log T). \tag{28}$$

Consequently,

$$\sum_{t=1}^{T}\widetilde{\ell}_t(\theta_t) - \min_{u\in\mathcal{K}}\sum_{t=1}^{T}\widetilde{\ell}_t(u) \le \frac{G_\lambda^2}{2\lambda}(1+\log T). \tag{29}$$

Proof. By Proposition 8.1, every loss $\widetilde{\ell}_t$ is λ-strongly convex and satisfies

$$\left\|\nabla\widetilde{\ell}_t(\theta_t)\right\|_2 \le G_\lambda.$$

Thus, the assumptions of Theorem 6.2 hold with $\alpha = \lambda$ and $G = G_\lambda$. Applying that theorem with $\eta_t = 1/(\lambda t)$ gives, for every $u \in \mathcal{K}$,

$$\sum_{t=1}^{T}\left(\widetilde{\ell}_t(\theta_t) - \widetilde{\ell}_t(u)\right) \le \frac{G_\lambda^2}{2\lambda}(1+\log T).$$

This proves Eq. (28). Taking u to be the best fixed regularized comparator proves Eq. (29). □

Dividing the preceding bound by T gives

$$\frac{1}{T}\left[\sum_{t=1}^{T}\widetilde{\ell}_t(\theta_t) - \min_{u\in\mathcal{K}}\sum_{t=1}^{T}\widetilde{\ell}_t(u)\right] \le \frac{G_\lambda^2}{2\lambda}\frac{1+\log T}{T} \xrightarrow[T\to\infty]{} 0.$$

Hence ridge OGD is no-regret for the regularized objective. Its average regularized loss becomes asymptotically no larger than that of the best fixed ridge coefficient selected in hindsight. Compared with the generic $O(\sqrt{T})$ bound for convex losses, strong convexity improves the cumulative rate to $O(\log T)$ and the average rate from $O(T^{-1/2})$ to $O(\log(T)/T)$ [4, 7].

What exactly is the hindsight comparator? Define

$$u_T^{\text{ridge}} \in \arg\min_{u\in\mathcal{K}}\sum_{t=1}^{T}\widetilde{\ell}_t(u).$$

For a fixed u,

$$\sum_{t=1}^{T} \widetilde{\ell}_t(u) = \frac{1}{2}\sum_{t=1}^{T}(x_t^\top u - y_t)^2 + \frac{T\lambda}{2}\|u\|_2^2 = T\left[\frac{1}{2T}\sum_{t=1}^{T}(x_t^\top u - y_t)^2 + \frac{\lambda}{2}\|u\|_2^2\right].$$

Multiplication by the positive constant T does not change the minimizer. Therefore,

$$u_T^{\text{ridge}} \in \arg\min_{u\in\mathcal{K}}\left\{\frac{1}{2T}\sum_{t=1}^{T}(x_t^\top u - y_t)^2 + \frac{\lambda}{2}\|u\|_2^2\right\}.$$

Adding the ridge penalty at every online round corresponds exactly to the usual batch ridge objective when the empirical squared loss is written as an average.

Remark 8.1 (The scaling convention matters)**.** If the intended batch objective is instead written with an *unnormalized* sum,

$$\frac{1}{2}\sum_{t=1}^{T}(x_t^\top u - y_t)^2 + \frac{\lambda}{2}\|u\|_2^2,$$

then the ridge penalty is paid only once. This is not the same objective as the one controlled by Theorem 8.1 with a fixed per-round penalty $\lambda\|\theta\|_2^2/2$.
For a known horizon, the corresponding per-round penalty would have to be scaled as $\lambda/(2T)$. Alternatively, FTRL is a natural formulation because it can add the regularizer once to the cumulative data-fitting loss. The value of λ must therefore always be interpreted together with the convention used for the empirical loss.

What ridge adds

Ridge does more than reduce the size of the coefficients. It shrinks directions that are weakly supported by the data, stabilizes the online updates, and adds uniform curvature to every per-round loss. This last property changes the available worst-case regret rate from $O(\sqrt{T})$ to $O(\log T)$.
The improvement has a modeling cost: the learner is now compared using a regularized objective, and larger values of λ produce stronger shrinkage. Ridge is therefore useful when stability and control of poorly identified directions matter, but it does not automatically improve the unregularized prediction error.

9 Detailed application III: pinball loss for inventory decisions

9.1 From forecast accuracy to operational cost

In inventory management, a prediction and a decision are not quite the same object. A point forecast attempts to describe future demand, whereas an order quantity must account for the economic consequences of ordering too much or too little. Two order quantities with the same absolute forecast error may therefore have very different operational costs.

At round t, let $q_t \in \mathcal{K} = [0, Q_{\max}]$ be the quantity ordered before demand y_t is observed. The upper bound $Q_{\max}$ may represent a storage, budget, or supply-capacity constraint. Once demand is revealed, two mutually exclusive situations can occur:

- if $q_t > y_t$, then $q_t - y_t$ units remain after demand has been served;
- if $q_t < y_t$, then $y_t - q_t$ units of demand cannot be served immediately.

Let $c_h > 0$ denote the marginal *holding cost* of one unit ordered in excess. Depending on the application, it may include storage, insurance, tied-up capital, depreciation, spoilage, or a future markdown. Let $c_b > 0$ denote the marginal *underage cost* of one unit of unmet demand. The notation c_b traditionally refers to a backorder cost, but it can more generally represent lost profit, emergency replenishment, late-delivery penalties, or deterioration of customer service.

These coefficients should be interpreted as effective costs per unit over one decision period. They need *not* correspond to a *single accounting entry*: they summarize the marginal operational consequences of surplus and shortage after terms that do not depend on q have been removed. With $a_+ = \max\{a, 0\}$, the resulting one-period loss is

$$\ell_t(q) = c_h(q - y_t)_+ + c_b(y_t - q)_+. \tag{30}$$

The two positive-part terms cannot be simultaneously positive. If $q > y_t$, only the holding term remains; if $q < y_t$, only the shortage term remains. Equivalently,

$$\ell_t(q) = \begin{cases} c_b(y_t - q), & q < y_t \\ 0, & q = y_t \\ c_h(q - y_t), & q > y_t. \end{cases}$$

This piecewise-linear construction is the simplest convex loss that assigns different marginal costs to underordering and overordering.

9.2 Connection with the pinball loss

Introduce the *critical level*

$$\tau = \frac{c_b}{c_b + c_h} \in (0, 1). \tag{31}$$

Recall that the pinball loss at level τ is

$$\rho_\tau(r) = \begin{cases} \tau r, & r \geq 0, \\ (\tau - 1)r, & r < 0. \end{cases}$$

Taking the residual to be $r = y_t - q$, we obtain

$$\begin{aligned}(c_b + c_h)\rho_\tau(y_t - q) &= \begin{cases} (c_b + c_h)\tau(y_t - q), & y_t - q \geq 0, \\ (c_b + c_h)(\tau - 1)(y_t - q), & y_t - q < 0 \end{cases} \\ &= \begin{cases} c_b(y_t - q), & q \leq y_t, \\ c_h(q - y_t), & q > y_t \end{cases} \\ &= c_h(q - y_t)_+ + c_b(y_t - q)_+.\end{aligned}$$

Consequently,

$$\ell_t(q) = (c_b + c_h)\,\rho_\tau(y_t - q). \tag{32}$$

The classical newsvendor loss is therefore exactly a scaled pinball loss. The scaling factor $c_b + c_h$ determines the overall monetary magnitude of the loss, while the ratio $\tau = c_b/(c_b + c_h)$ determines its asymmetry. This identity connects inventory optimization with quantile forecasting and quantile regression [1, 3, 6].

Example 9.1 (Why the relative costs matter)**.** Let $c_b = 3$ and $c_h = 1$. An unmet unit is then three times as costly as an excess unit, and

$$\tau = \frac{3}{3+1} = \frac{3}{4}.$$

- If $q_t = 13$ and $y_t = 16$, three units of demand are unmet, so $\ell_t(13) = 3(16 - 13) = 9$.
- If instead $q_t = 16$ and $y_t = 14$, two units remain, so $\ell_t(16) = 1(16 - 14) = 2$.

Underordering is more expensive than overordering. The order quantity should therefore be placed above the median of the demand distribution: the relevant target is its 75% quantile.

More generally, if $c_b = c_h$, then $\tau = 1/2$ and the loss targets a median. Increasing c_b raises τ and produces a more conservative, higher order quantity. Increasing c_h lowers τ and discourages excess inventory.

9.3 The optimal order quantity is a critical quantile

The critical ratio in (31) is not an arbitrary modeling convention. It follows from the first-order optimality condition for the expected inventory cost. The next proposition gives the precise statement, including distributions with point masses.

Proposition 9.1 (Critical quantile for asymmetric inventory costs)**.** *Let Y be an integrable real-valued demand variable with cumulative distribution function*

$$F : q \mapsto \mathbb{P}(Y \leq q),$$

and for $c_h > 0$, $c_b > 0$, let

$$C(q) = \mathbb{E}[c_h(q - Y)_+ + c_b(Y - q)_+].$$

Define $\tau = \frac{c_b}{c_b + c_h}$. Then $q^\star \in \mathbb{R}$ minimizes C if and only if

$$F(q^{\star -}) \leq \tau \leq F(q^\star), \tag{33}$$

where $F(q^-) = \mathbb{P}(Y < q)$ is the left limit of F at q.

Thus every optimal order quantity is a τ-quantile of demand. If F is continuous and strictly increasing, the minimizer is unique and satisfies

$$q^\star = F^{-1}(\tau) = F^{-1}\left(\frac{c_b}{c_b + c_h}\right).$$

Proof. For a fixed realization y, define

$$\ell(q; y) = c_h(q - y)_+ + c_b(y - q)_+.$$

This function is convex and piecewise linear in q. Its slope is $-c_b$ to the left of y and c_h to the right of y, while at $q = y$ its subdifferential is the interval $[-c_b, c_h]$:

$$\partial_q \ell(q; y) = \begin{cases} \{-c_b\}, & q < y, \\ [-c_b, c_h], & q = y, \\ \{c_h\}, & q > y. \end{cases}$$

To account correctly for a possible point mass at q, consider the left and right derivatives of the expected cost. For the left derivative, observations satisfying $Y = q$ contribute the slope $-c_b$, and hence

$$C'_-(q) = c_h\,\mathbb{P}(Y < q) - c_b\,\mathbb{P}(Y \geq q) = c_h F(q^-) - c_b(1 - F(q^-)) = (c_b + c_h)F(q^-) - c_b.$$

For the right derivative, observations satisfying $Y = q$ contribute the slope c_h, so

$$C'_+(q) = c_h\,\mathbb{P}(Y \leq q) - c_b\,\mathbb{P}(Y > q) = c_h F(q) - c_b(1 - F(q)) = (c_b + c_h)F(q) - c_b.$$

Because C is convex, according to the statement of Eq. (4), $q^\star$ is a minimizer if and only if

$$C'_-(q^\star) \leq 0 \leq C'_+(q^\star).$$

Substituting the two expressions above gives

$$(c_b + c_h)F(q^{\star-}) - c_b \leq 0 \leq (c_b + c_h)F(q^\star) - c_b.$$

Since $c_b + c_h > 0$, division by this quantity yields

$$F(q^{\star-}) \leq \frac{c_b}{c_b + c_h} \leq F(q^\star),$$

which is precisely the quantile condition (33). □

For a finite demand sequence $y_1, \ldots, y_T$, the same argument applies to the empirical distribution

$$\widehat{F}_T(q) = \frac{1}{T}\sum_{t=1}^{T} \mathbb{1}_{\{y_t \leq q\}}.$$

Consequently, any minimizer of

$$\sum_{t=1}^{T} [c_h(q - y_t)_+ + c_b(y_t - q)_+]$$

is an empirical τ-quantile, with $\tau = c_b/(c_b + c_h)$. This is the finite-sample counterpart of the classical critical-fractile formula [6, 3].

If the order quantity is restricted to $[0, Q_{\max}]$ and an empirical τ-quantile belongs to that interval, it remains optimal. If all unconstrained minimizers lie below or above the feasible interval, the constrained minimizer is respectively 0 or $Q_{\max}$.

9.4 The hindsight comparator has an operational meaning

In the online problem, no probability distribution for demand needs to be specified. After observing the entire deterministic sequence $y_1, \ldots, y_T$, the natural comparator is the best single feasible order quantity:

$$q_T^\star \in \operatorname*{arg\,min}_{q \in [0, Q_{\max}]} \sum_{t=1}^{T} \ell_t(q).$$

By Proposition 9.1, $q_T^\star$ is a capacity-feasible empirical τ-quantile. The corresponding inventory regret is

$$\mathrm{Reg}_T^{\mathrm{inv}} = \sum_{t=1}^{T} \ell_t(q_t) - \min_{q \in [0, Q_{\max}]} \sum_{t=1}^{T} \ell_t(q). \tag{34}$$

This quantity compares the sequential policy with an oracle that sees the full demand sequence before selecting its order quantity. The oracle has informational hindsight, but it is still required to use the same quantity on every round. The comparison is therefore demanding without being unattainably flexible.

9.5 The online update is an operational correction rule

The inventory loss is not differentiable at $q = y_t$, but projected OGD only requires a subgradient. At the played order quantity q_t, one may select

$$g_t \in \partial \ell_t(q_t) = \begin{cases} -c_b, & q_t < y_t \\ [-c_b, c_h], & q_t = y_t \\ c_h, & q_t > y_t. \end{cases} \tag{35}$$

The projected OGD update is therefore

$$q_{t+1} = \Pi_{[0,Q_{\max}]}(q_t - \eta_t g_t)\,. \tag{36}$$

This update has a direct operational interpretation.

- After a shortage, $q_t < y_t$ and $g_t = -c_b$, so $q_{t+1} = \Pi_{[0,Q_{\max}]}(q_t + \eta_t c_b)$. The next order quantity increases.
- After excess inventory, $q_t > y_t$ and $g_t = c_h$, so $q_{t+1} = \Pi_{[0,Q_{\max}]}(q_t - \eta_t c_h)$. The next quantity decreases.
- At an exact match, $0 \in [-c_b, c_h]$, and choosing $g_t = 0$ leaves the order quantity unchanged.

The magnitude of the prediction error does not appear in the subgradient because the loss is linear on either side of y_t: each additional unit of shortage costs the same amount c_b, and each additional unit of surplus costs the same amount c_h. Nevertheless, the costs determine both the target quantile and the relative sizes of the upward and downward updates.

9.6 Regret guarantee for the inventory policy

For every t and every q_t, the subgradient in (35) satisfies $|g_t| \le G$, with $G := \max\{c_b, c_h\}$. Moreover, the feasible interval $[0, Q_{\max}]$ has Euclidean diameter $D = Q_{\max}$. Applying Theorem 6.1 with a constant learning rate $\eta > 0$ gives

$$\mathrm{Reg}_T^{\mathrm{inv}} \le \frac{Q_{\max}^2}{2\eta} + \frac{\eta G^2 T}{2}. \tag{37}$$

The first term penalizes a learning rate that is too small, because the algorithm then moves too slowly from its initial quantity. The second term penalizes a learning rate that is too large, because each subgradient produces a stronger and potentially more volatile correction.

Balancing the two terms gives $\eta = Q_{\max}/(G\sqrt{T})$, and therefore

$$\mathrm{Reg}_T^{\mathrm{inv}} \le Q_{\max} G \sqrt{T}. \tag{38}$$

Consequently,

$$\frac{\mathrm{Reg}_T^{\mathrm{inv}}}{T} \le \frac{Q_{\max} G}{\sqrt{T}} \xrightarrow[T\to\infty]{} 0.$$

Thus no positive asymptotic average-cost gap remains between the online policy and the best fixed feasible order quantity in hindsight. This guarantee does not require demands to be independent, identically distributed, or stationary; it holds for every demand sequence covered by the online protocol.

9.7 Extension to order quantities based on features

A single scalar quantity cannot react to observable information such as price, promotions, seasonality, or day of the week. When a feature vector $x_t \in \mathbb{R}^d$ is available before ordering, one may instead predict

$$q_t = x_t^\top \theta_t,$$

with feasibility enforced by the parameter set, a projection, or an appropriate link function. If $s_t \in \partial_q \ell_t(q_t)$ denotes the scalar slope in (35), then a subgradient with respect to the parameter vector is

$$g_t^\theta = s_t x_t.$$

The resulting update learns a conditional quantile rather than one unconditional order quantity. It is therefore an online form of quantile regression: the features determine how the order quantity changes with the observed context, while the ratio $c_b/(c_b + c_h)$ continues to encode the operational asymmetry [1].

What to remember about pinball loss

The pinball loss converts unequal inventory costs into a quantile target:

$$\tau = \frac{c_b}{c_b + c_h}.$$

A larger shortage cost c_b raises τ and leads to higher order quantities, whereas a larger holding cost c_h lowers τ. OGD then increases the order after a shortage and decreases it after excess inventory. Although the loss is only convex and yields an $O(\sqrt{T})$ regret guarantee, this guarantee concerns the operational cost that the store actually seeks to control.

From operating costs to an online algorithm

This section converts an operational asymmetry into a complete mathematical learning problem.

- The marginal shortage and holding costs define a convex pinball loss;
- their ratio selects the economically relevant demand quantile;
- the subgradient turns each shortage or surplus into an explicit correction;
- and OGD controls the cumulative cost relative to the best fixed feasible quantity.

The resulting guarantee is distribution-free, but its comparator remains static: tracking a different optimal quantity at every round would require a dynamic-regret analysis.

10 What the regret guarantee does – and does not – say

10.1 Deterministic guarantee versus statistical guarantee

The OGD regret bound is pathwise: once the assumptions hold, it is true for every admissible sequence of convex losses. There is no expectation and no confidence level. This robustness is one of the central attractions of online convex optimization.

However, regret alone does not directly provide a confidence interval, a generalization probability, or a causal interpretation. Under additional stochastic assumptions, online-to-batch conversion can turn an online regret bound into an excess-risk bound for an averaged iterate [7]. Conceptually, if

Z_t are identically distributed and $\ell(\theta; Z_t)$ is convex, then one can use

$$\bar{\theta}_T = \frac{1}{T}\sum_{t=1}^{T}\theta_t$$

and convexity to relate its population risk to average online performance. The stochastic assumptions enter at this conversion step, not in the original regret proof.

10.2 Common logical mistakes

The definitions above are compact, which makes several distinct errors look deceptively similar. The following checklist is best read as a set of questions to ask before trusting either a proof or an implementation.

1. **Using y_t before predicting at time t.** This is leakage and destroys the online protocol.
2. **Calling decreasing training loss a regret guarantee.** Regret requires an explicit comparator evaluated on the same loss sequence.
3. **Forgetting bounded gradients.** Squared loss has unbounded gradients on $\mathbb{R}^d$; a domain or iterate-control argument is needed.
4. **Confusing a regularizer in the algorithm with a regularizer in every loss.** FTRL regularization and per-round penalized losses lead to different comparators.
5. **Interpreting no-regret as parameter convergence.** Regret controls cumulative values, not necessarily the trajectory itself.
6. **Comparing against an unrestricted moving oracle.** Without a variation constraint, such a benchmark can fit every round and makes meaningful tracking impossible.
7. **Using the wrong quantile for inventory.** With the convention in this chapter, the target is $\tau = c_b/(c_b + c_h)$; a larger backorder cost must increase, not decrease, the target order quantity.

10.3 Choosing an algorithm from the geometry

There is no universally best online method. A useful choice begins with the feasible set, the regularity of the loss, and the kind of structure the update must preserve; the table summarizes those links rather than ranking the algorithms.

Situation	Natural method	Reason
Bounded Euclidean parameter vector	Projected OGD	Simple update and $O(\sqrt{T})$ regret
Probability vector on a simplex	Entropic OMD / Hedge	Preserves positivity and unit sum
Very different coordinate scales	AdaGrad	Coordinate-adaptive steps
Composite smooth loss plus ℓ_1 penalty	Proximal OGD or FTRL	Handles nonsmooth sparsity exactly
Asymmetric inventory cost	Scalar projected OGD or FTRL	Learns the cost-determined quantile
Strongly convex per-round losses	OGD with $\eta_t \propto 1/t$	Logarithmic regret
Changing comparator	Dynamic-regret method	Explicitly controls tracking difficulty

Table 1: Typical online-learning methods for different problem structures.

Conclusion

Online learning is a theory of decision-making under chronology. At each round, the learner must choose a decision before observing the corresponding outcome or loss, and may use the resulting feedback only to prepare future decisions. The objective is therefore not simply to fit one model on a completed dataset, but to control the performance of an entire sequence of decisions.

Regret provides the mathematical criterion for this objective. It compares the learner's cumulative loss with that of a clearly specified hindsight comparator. In particular, a *sublinear static-regret bound* shows that the additional loss paid for acting sequentially *becomes negligible on average*. This guarantee is distribution-free: it can hold for arbitrary admissible loss sequences, without independence or stationarity assumptions. Its interpretation must nevertheless remain precise: small static regret guarantees competition with the best fixed decision, but does not by itself imply convergence of the iterates, probabilistic generalization, or successful tracking of a moving optimum.

Online Gradient Descent and regret therefore play different but complementary roles. Regret defines what successful online performance means, whereas OGD is an algorithm designed to achieve it. Its update

$$\theta_{t+1} = \Pi_{\mathcal{K}}(\theta_t - \eta_t g_t)$$

uses only the current decision and the newly observed subgradient. The one-step potential inequality then connects this local correction to a global cumulative guarantee. For general convex losses with bounded subgradients, this argument gives $O(\sqrt{T})$ regret; when every per-round loss has uniform strong convexity, the additional curvature allows a decreasing learning rate and an $O(\log T)$ bound. Thus, the analysis explains not only whether an algorithm works, but also which properties of the loss make faster guarantees possible. In both cases, the average regret *vanishes*, meaning that, in the long run, *OGD's average loss becomes as small as that of the best fixed decision selected in hindsight* by an "oracle" that has observed the entire loss sequence.

The other methods studied in this chapter modify the same general principle according to the structure of the problem. FTL directly optimizes past losses but may be unstable; FTRL adds regularization to control this instability; mirror descent adapts the update to a non-Euclidean geometry; and AdaGrad adjusts the learning rate to the observed scale of each coordinate. Ridge supplies shrinkage and strong convexity, while lasso promotes sparse parameter vectors through a nonsmooth penalty. These methods should therefore be understood as different choices of stability, geometry, and adaptation rather than as unrelated algorithmic recipes.

The running supermarket example illustrates how these mathematical choices correspond to different practical objectives. Squared-loss OGD learns a sequential demand predictor; ridge stabilizes its coefficients; and pinball loss translates holding and backorder costs into a cost-sensitive order quantity targeting the appropriate demand quantile. In each case, the loss specifies what constitutes an error, the algorithm specifies how feedback changes the next decision, and the regret bound certifies the cumulative price of learning sequentially.

The central lesson is therefore that online learning separates three questions that must not be confused: *what decision is feasible*, *what loss represents the objective*, and *which update can control cumulative performance*. Once these elements are specified, regret analysis provides a rigorous bridge from local, implementable updates to finite-time guarantees for sequential decision-making.

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Appendix: inequalities used in the convergence proof

This appendix proves the two inequalities used in the convergence analysis of gradient descent. The assumptions of smoothness and strong convexity are stated separately because they play different roles: smoothness controls the decrease produced by one gradient step, whereas strong convexity relates the gradient norm to the distance from the minimum.

Proposition 10.1 (Descent lemma). *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and L-smooth, meaning that*

$$\|\nabla f(u) - \nabla f(v)\|_2 \le L \|u - v\|_2 \qquad \text{for all } u, v \in \mathbb{R}^d.$$

Then, for every $u, v \in \mathbb{R}^d$,

$$f(v) \le f(u) + \langle \nabla f(u), v - u \rangle + \frac{L}{2} \|v - u\|_2^2 . \tag{39}$$

Proof. Fix $u, v \in \mathbb{R}^d$ and set $d = v - u$. Consider the one-dimensional function

$$\varphi(s) = f(u + sd), \qquad s \in [0, 1].$$

By the chain rule,

$$\varphi'(s) = \langle \nabla f(u + sd), d \rangle .$$

The fundamental theorem of calculus gives

$$\begin{aligned} f(v) - f(u) &= \varphi(1) - \varphi(0) \\ &= \int_0^1 \langle \nabla f(u + sd), d \rangle \, \mathrm{d}s \\ &= \langle \nabla f(u), d \rangle + \int_0^1 \langle \nabla f(u + sd) - \nabla f(u), d \rangle \, \mathrm{d}s. \end{aligned}$$

By the Cauchy–Schwarz inequality and L-smoothness,

$$\langle \nabla f(u + sd) - \nabla f(u), d \rangle \le \|\nabla f(u + sd) - \nabla f(u)\|_2 \, \|d\|_2 \le Ls \, \|d\|_2^2 .$$

Therefore,

$$f(v) - f(u) \le \langle \nabla f(u), d \rangle + \int_0^1 Ls \, \|d\|_2^2 \, \mathrm{d}s = \langle \nabla f(u), d \rangle + \frac{L}{2} \|d\|_2^2 .$$

Substituting $d = v - u$ proves (39). □

Proposition 10.2 (Decrease produced by one gradient step). *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and L-smooth. For $\theta \in \mathbb{R}^d$ and $\eta > 0$, define*

$$\theta^+ = \theta - \eta \nabla f(\theta).$$

Then

$$f(\theta^+) \le f(\theta) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla f(\theta)\|_2^2 . \tag{40}$$

Proof. Apply the descent lemma, Proposition 10.1, with

$$u = \theta, \qquad v = \theta^+ = \theta - \eta \nabla f(\theta).$$

Since

$$\theta^+ - \theta = -\eta \nabla f(\theta),$$

we obtain

$$\begin{aligned} f(\theta^+) &\le f(\theta) + \langle \nabla f(\theta), -\eta \nabla f(\theta) \rangle + \frac{L}{2} \left\| -\eta \nabla f(\theta) \right\|_2^2 \\ &= f(\theta) - \eta \left\| \nabla f(\theta) \right\|_2^2 + \frac{L\eta^2}{2} \left\| \nabla f(\theta) \right\|_2^2 \\ &= f(\theta) - \eta \left(1 - \frac{L\eta}{2} \right) \left\| \nabla f(\theta) \right\|_2^2 . \end{aligned}$$

This proves (40).

□

Proposition 10.3 (Consequences of strong convexity)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and μ-strongly convex for some $\mu > 0$, meaning that*

$$f(v) \ge f(u) + \langle \nabla f(u), v - u \rangle + \frac{\mu}{2} \left\| v - u \right\|_2^2$$

for every $u, v \in \mathbb{R}^d$. Then f has a unique minimizer $\theta^\star$. Moreover, for every $\theta \in \mathbb{R}^d$,

$$\left\| \nabla f(\theta) \right\|_2^2 \ge 2\mu \big(f(\theta) - f(\theta^\star) \big), \tag{41}$$

and

$$f(\theta) - f(\theta^\star) \ge \frac{\mu}{2} \left\| \theta - \theta^\star \right\|_2^2 . \tag{42}$$

Proof. We first establish existence and uniqueness. Strong convexity applied with $u = 0$ and $v = \theta$ gives

$$f(\theta) \ge f(0) + \langle \nabla f(0), \theta \rangle + \frac{\mu}{2} \left\| \theta \right\|_2^2 .$$

By Cauchy–Schwarz,

$$f(\theta) \ge f(0) - \left\| \nabla f(0) \right\|_2 \left\| \theta \right\|_2 + \frac{\mu}{2} \left\| \theta \right\|_2^2 .$$

The quadratic term dominates the linear term as $\left\| \theta \right\|_2 \to \infty$. Hence f is coercive. Since f is differentiable and therefore continuous, it attains its minimum at some point $\theta^\star$.
Because the optimization problem is unconstrained,

$$\nabla f(\theta^\star) = 0.$$

If there were another minimizer $v \neq \theta^\star$, strong convexity would give

$$f(v) \ge f(\theta^\star) + \langle \nabla f(\theta^\star), v - \theta^\star \rangle + \frac{\mu}{2} \left\| v - \theta^\star \right\|_2^2 = f(\theta^\star) + \frac{\mu}{2} \left\| v - \theta^\star \right\|_2^2 > f(\theta^\star),$$

which contradicts the fact that v is also a minimizer. Thus $\theta^\star$ is unique.
We now prove the gradient bound. Apply strong convexity with $u = \theta$ and $v = \theta^\star$:

$$f(\theta^\star) \ge f(\theta) + \langle \nabla f(\theta), \theta^\star - \theta \rangle + \frac{\mu}{2} \left\| \theta^\star - \theta \right\|_2^2 .$$

Rearranging gives

$$f(\theta) - f(\theta^\star) \le \langle \nabla f(\theta), \theta - \theta^\star \rangle - \frac{\mu}{2} \|\theta - \theta^\star\|_2^2 .$$

Set

$$g = \nabla f(\theta), \qquad d = \theta - \theta^\star.$$

Completing the square yields

$$\begin{aligned} \langle g, d \rangle - \frac{\mu}{2} \|d\|_2^2 &= \frac{1}{2\mu} \|g\|_2^2 - \frac{\mu}{2} \left\| d - \frac{g}{\mu} \right\|_2^2 \\ &\le \frac{1}{2\mu} \|g\|_2^2 . \end{aligned}$$

Therefore,

$$f(\theta) - f(\theta^\star) \le \frac{1}{2\mu} \|\nabla f(\theta)\|_2^2 ,$$

which is equivalent to (41).
Finally, apply strong convexity with $u = \theta^\star$ and $v = \theta$. Since $\nabla f(\theta^\star) = 0$,

$$\begin{aligned} f(\theta) &\ge f(\theta^\star) + \langle \nabla f(\theta^\star), \theta - \theta^\star \rangle + \frac{\mu}{2} \|\theta - \theta^\star\|_2^2 \\ &= f(\theta^\star) + \frac{\mu}{2} \|\theta - \theta^\star\|_2^2 . \end{aligned}$$

This proves (42). □